A1: RESEARCH QUESTION

What features determine the Monthly Charges in the Churn dataset?

A2: GOALS

The main goal of the data analysis is to determine what features are important in determining the amount of the monthly charge for our customers.

According to Bhandari, Pritha (2023, June 22), the independent variable is the cause. In contrast, the dependent variable is the effect, and its value depends on the changes of the independent variable in a cause-effect relationship. This analysis aims to determine which features significantly impact the monthly charge using the Linear regression analysis.

B1: SUMMARY OF ASSUMPTIONS

Linear regression is a method to understand the relationship between the multiple explanatory or predictor variables and the response variable and is used when the response variable is numerical.

According to Bobbitt, Zach (2021, November 16), Multiple Linear regression has the following assumptions

1. Linear relationship

   A linear relationship exists between each predictor variable and the response variable.

2. No Multicollinearity

   None of the predictor variables are highly correlated with each other.

3. Independence of the observations.
4. Homoscedasticity.

The residuals have constant variance at every point in the Linear model.

5. Multivariate normality

   The residuals of the model are normally distributed

B2: TOOL BENEFITS

For the purpose of undertaking the Linear regression analysis, I will be utilizing the Python tool.

Python is an open-source general-purpose programming language with many packages that can be used and shared with others. I utilized Python Programming Language because of its flexibility and extensive library collections, which are valuable for performing complex calculations and Linear regression analysis.

The following libraries and packages will be utilized

- Numpy

  Numpy will be utilized for basic operations

- Pandas

  Pandas will be used to import data into the dataframe, observe the data in the dataframe,

- Scipy

  Scipy will be utilized for statistics

- Statsmodels

  Statsmodels will be utilized for the computation of linear regression.

- Matplotlib

  Matplotlib will be utilized for visualization.

- Seaborn

Seaborn will be utilized for visualization.

- Scikit -learn

  Sklearn will be utilized for one hot encoding.

The libraries will be used to import data into the dataframe, observe the data in the dataframe, perform statistics, convert categorical data to numerical data, create the visualization, and perform linear regression analysis.

B3: APPROPRIATE TECHNIQUE

According to Coursera staff (2024, June 27), Regression analysis is a method that allows us to understand the relationship between two or more variables. Linear regression is a specific type of regression analysis used when you expect a straight-line relationship between the independent and dependent variables.

Linear regression is a method to understand the relationship between the multiple explanatory or predictor variables and the response variable and is used when the response variable is numerical. In this case, the response variable we are trying to predict is the Monthly charge, which has a numerical data type. This analysis will establish the linear relationship between the monthly charge (response variable) and the predictor variables.

Since the analysis aims to determine the significant features that impact the monthly charge, and based on the dataset, we have multiple independent variables. Multiple linear regression analysis is the ideal technique. More so, to be able to analyze all the variables, the categorical data will be converted to numerical data since multiple regression analysis analyses the relationship between quantitative or numerical data.

C1: DATA CLEANING

After loading the data, it was viewed and reviewed to understand its nature.

1. Data type constraint and Data Missingness

   Due to the wrong datatype specified while loading the Internet service, the value of Other was identified as missing. To address the issue, the data type was specified as missing data. To address the issue, the value feature was specified as a String

2. Data Uniqueness

   The data was assessed to determine if there were any duplicate values. No duplicate values were found in both the columns and the rows.

3. Drop redundant features with high cardinality and features with high collinearity. The features of CaseOrder, Customer_id, Interaction, UID, City, State, County, Zip, Lat, Lng, Job, TimeZone, Churn, Bandwidth_GB_Year, Tenure, InternetService_Other, TimelyResponse and Fixes

4. Check for Outliers. I will find Outliers in the data set by plotting to identify outliers or outlying observations. Due to lack of understanding of the domain of the data provided/other pertinent information with regards to the data , the features with outliers were retained.

5. Convert Categorical data to numerical data through dummy encoding (One Hot encoding)

   According to Saxena and Shipra (2024, September 18), preprocessing categorical data is necessary since most machine learning models only accept numerical data. One way to do this is through one hot encoding, which is used when the features are nominal, and each category is mapped with a binary variable 0 or 1. 0 represents absence, and 1 represents the presence of that category. The newly created binary features are known as Dummy variables. Dummy encoding uses N-1 features to represent N labels/categories.

C2: SUMMARY STATISTICS

1. MonthlyCharge

The data type for the monthly charge feature is numerical. The monthly Charge will be used as a dependent variable.

```
count    10000.000000
mean       172.624816
std         42.943094
min         79.978860
25%        139.979239
50%        167.484700
75%        200.734725
max        290.160419
Name: MonthlyCharge, dtype: float64
```

2. Area

The data type for the Area variable is categorical. The area variable will be used as an independent variable. The variable will be converted to numerical data.

```
count        10000
unique           3
top       Suburban
freq          3346
Name: Area, dtype: object
['Urban' 'Suburban' 'Rural']
```

3. Children

The data type for the Children feature is numerical. The children variable will be used as an independent variable.

```
count    10000.0000
mean         2.0877
std          2.1472
min          0.0000
25%          0.0000
50%          1.0000
75%          3.0000
max         10.0000
Name: Children, dtype: float64
```

4. Age

The data type for the Age feature is numerical. The Age variable will be used as an independent variable.

```
count    10000.000000
mean        53.078400
std         20.698882
min         18.000000
25%         35.000000
50%         53.000000
75%         71.000000
max         89.000000
Name: Age, dtype: float64
```

5. Income

The data type for the income feature is numerical. The income variable will be used as an independent variable.

```
count     10000.000000
mean      39806.926771
std       28199.916702
min         348.670000
25%       19224.717500
50%       33170.605000
75%       53246.170000
max      258900.700000
Name: Income, dtype: float64
```

6. Marital

The data type for the Marital variable is categorical. The Marital variable will be used as an independent variable. The variable will be converted to numerical data.

```
count          10000
unique             5
top         Divorced
freq            2092
Name: Marital, dtype: object
['Widowed' 'Married' 'Separated' 'Never Married' 'Divorced']
```

7. Gender

The data type for the Gender variable is categorical. The Gender variable will be used as an independent variable. The variable will be converted to numerical data.

```
count        10000
unique           3
top         Female
freq          5025
Name: Gender, dtype: object
['Male' 'Female' 'Nonbinary']
```

8. Outage_sec_perweek

The data type for the Outage_sec_perweek feature is numerical. The Outage_sec_perweek variable will be used as an independent variable.

```
count    10000.000000
mean        10.001848
std          2.976019
min          0.099747
25%          8.018214
50%         10.018560
75%         11.969485
max         21.207230
Name: Outage_sec_perweek, dtype: float64
```

9. Email

The data type for the Email feature is numerical. The Email variable will be used as an independent variable.

```
count    10000.000000
mean        12.016000
std          3.025898
min          1.000000
25%         10.000000
50%         12.000000
75%         14.000000
max         23.000000
Name: Email, dtype: float64
```

10. Contacts

The data type for the contacts feature is numerical. The contact variable will be used as an independent variable.

```
count    10000.000000
mean         0.994200
std          0.988466
min          0.000000
25%          0.000000
50%          1.000000
75%          2.000000
max          7.000000
Name: Contacts, dtype: float64
```

11. Yearly_equip_failure

The data type for the Yearly_equip_failure feature is numerical. The yearly_equip_failure variable will be used as an independent variable.

```
count    10000.000000
mean         0.398000
std          0.635953
min          0.000000
25%          0.000000
50%          0.000000
75%          1.000000
max          6.000000
Name: Yearly_equip_failure, dtype: float64
```

12. Techie

The data type for the Techie variable is categorical. The Techie variable will be used as an independent variable. The variable will be converted to numerical data.

```
count     10000
unique        2
top          No
freq       8321
Name: Techie, dtype: object
['No' 'Yes']
```

13. Contract

The data type for the Contract variable is categorical. The contract variable will be used as an independent variable. The variable will be converted to numerical data.

```
count    10000.000000
mean         0.994200
std          0.988466
min          0.000000
25%          0.000000
50%          1.000000
75%          2.000000
max          7.000000
Name: Contacts, dtype: float64
```

14. Port_modem

The data type for the Port_modem variable is categorical. The Port_modem variable will be used as an independent variable. The variable will be converted to numerical data.

```
count     10000
unique        2
top          No
freq       5166
Name: Port_modem, dtype: object
['Yes' 'No']
```

15. Tablet

The data type for the Tablet variable is categorical. The Tablet variable will be used as an independent variable. The variable will be converted to numerical data.

```
count     10000
unique        2
top          No
freq       7009
Name: Tablet, dtype: object
['Yes' 'No']
```

16. InternetService

The data type for the InternetService variable is categorical. The InternetService variable will be used as an independent variable. The variable will be converted to numerical data.

```
count            10000
unique               3
top        Fiber Optic
freq              4408
Name: InternetService, dtype: object
['Fiber Optic' 'DSL' 'Other ']
```

17. Phone

The data type for the Phone variable is categorical. The Phone variable will be used as an independent variable. The variable will be converted to numerical data.

```
count     10000
unique        2
top         Yes
freq       9067
Name: Phone, dtype: object
['Yes' 'No']
```

18. Multiple

The data type for the Multiple variables is categorical. Multiple variables will be used as independent variables. The variable will be converted to numerical data.

```
count     10000
unique        2
top          No
freq       5392
Name: Multiple, dtype: object
['No' 'Yes']
```

19. OnlineSecurity

The data type for the Online Security variable is categorical. The Online Security variable will be used as an independent variable. The variable will be converted to numerical data.

```
count     10000
unique        2
top          No
freq       6424
Name: OnlineSecurity, dtype: object
['Yes' 'No']
```

20. OnlineBackup

The data type for the Online Backup variable is categorical. The Online Backup variable will be used as an independent variable. The variable will be converted to numerical data.

```
count     10000
unique        2
top          No
freq       5494
Name: OnlineBackup, dtype: object
['Yes' 'No']
```

21. DeviceProtection

The data type for the Device Protection variable is categorical. The Device Protection variable will be used as an independent variable. The variable will be converted to numerical data.

```
count     10000
unique        2
top          No
freq       5614
Name: DeviceProtection, dtype: object
['No' 'Yes']
```

22. TechSupport

The data type for the TechSupport variable is categorical. The TechSupport variable will be used as an independent variable. The variable will be converted to numerical data.

```
count     10000
unique        2
top          No
freq       6250
Name: TechSupport, dtype: object
['No' 'Yes']
```

23. StreamingTV

The data type for the StreamingTV variable is categorical. The StreamingTV variable will be used as an independent variable.

```
count     10000
unique        2
top          No
freq       5071
Name: StreamingTV, dtype: object
['No' 'Yes']
```

24. StreamingMovies

The data type for the Streaming Movies variable is categorical. The Streaming Movies variable will be used as an independent variable. The variable will be converted to numerical data.

```
count     10000
unique        2
top          No
freq       5110
Name: StreamingMovies, dtype: object
['Yes' 'No']
```

25. PaperlessBilling

The data type for the Paperless Billing variable is categorical. The PaperlessBilling variable will be used as an independent variable. The variable will be converted to numerical data.

```
count     10000
unique        2
top         Yes
freq       5882
Name: PaperlessBilling, dtype: object
['Yes' 'No']
```

26. PaymentMethod

The data type for the PaymentMethod variable is categorical. The Payment Method variable will be used as an independent variable. The variable will be converted to numerical data.

```
count                 10000
unique                    4
top        Electronic Check
freq                   3398
Name: PaymentMethod, dtype: object
['Credit Card (automatic)' 'Bank Transfer(automatic)' 'Mailed Check'
 'Electronic Check']
```

27. Population

The data type for the Population feature is numerical. The population variable will be used as an independent variable.

```
count     10000.000000
mean       9756.562400
std       14432.698671
min           0.000000
25%         738.000000
50%        2910.500000
75%       13168.000000
max      111850.000000
Name: Population, dtype: float64
```

28. Replacements

The data type for the Replacements variable is numerical. The Replacements variable will be used as an independent variable.

```
count    10000.000000
mean         3.487000
std          1.027977
min          1.000000
25%          3.000000
50%          3.000000
75%          4.000000
max          8.000000
Name: Replacements, dtype: float64
```

29. Reliability

The data type for the Reliability variable is numerical. The Reliability variable will be used as an independent variable.

```
count    10000.000000
mean         3.497500
std          1.025816
min          1.000000
25%          3.000000
50%          3.000000
75%          4.000000
max          7.000000
Name: Reliability, dtype: float64
```

30. Options

The data type for the Options variable is numerical. The Options variable will be used as an independent variable.

```
count    10000.000000
mean         3.492900
std          1.024819
min          1.000000
25%          3.000000
50%          3.000000
75%          4.000000
max          7.000000
Name: Options, dtype: float64
```

31. Respectfulness

The data type for the Respectfulness variable is numerical. The Respectfulness variable will be used as an independent variable.

```
count    10000.000000
mean         3.497300
std          1.033586
min          1.000000
25%          3.000000
50%          3.000000
75%          4.000000
max          8.000000
Name: Respectfulness, dtype: float64
```

32. Courteous

The data type for the Courteous variable is numerical. The Courteous variable will be used as an independent variable.

```
count    10000.000000
mean         3.509500
std          1.028502
min          1.000000
25%          3.000000
50%          4.000000
75%          4.000000
max          7.000000
Name: Courteous, dtype: float64
```

33. Listening

The data type for the Listening variable is numerical. The Listening variable will be used as an independent variable.

```
count    10000.000000
mean         3.495600
std          1.028633
min          1.000000
25%          3.000000
50%          3.000000
75%          4.000000
max          8.000000
Name: Listening, dtype: float64
```

```
Churn_df_encoded.info()
```

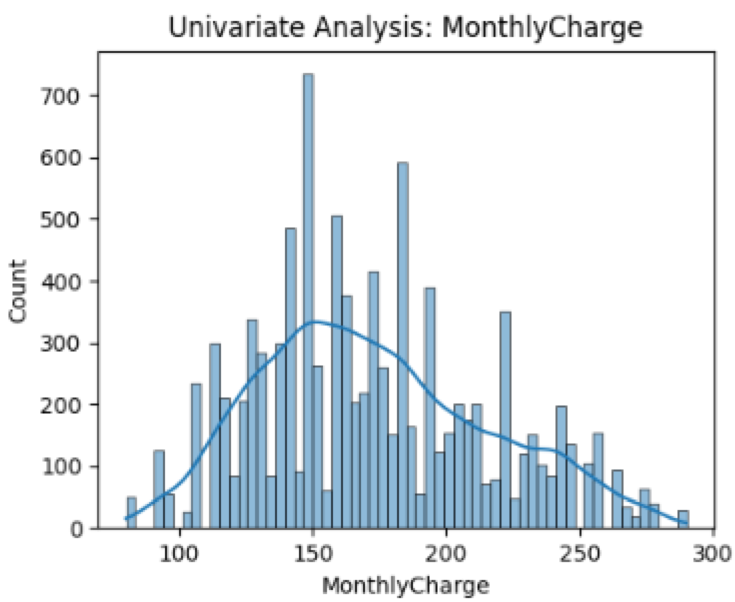
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 41 columns):
 #   Column                               Non-Null Count  Dtype  
---  ------                               --------------  -----  
 0   Population                           10000 non-null  int64  
 1   Children                             10000 non-null  int64  
 2   Age                                  10000 non-null  int64  
 3   Income                               10000 non-null  float64
 4   Outage_sec_perweek                   10000 non-null  float64
 5   Email                                10000 non-null  int64  
 6   Contacts                             10000 non-null  int64  
 7   Yearly_equip_failure                 10000 non-null  int64  
 8   MonthlyCharge                        10000 non-null  float64
 9   Replacements                         10000 non-null  int64  
 10  Reliability                          10000 non-null  int64  
 11  Options                              10000 non-null  int64  
 12  Respectfulness                       10000 non-null  int64  
 13  Courteous                            10000 non-null  int64  
 14  Listening                            10000 non-null  int64  
 15  Area_Suburban                        10000 non-null  int64  
 16  Area_Urban                           10000 non-null  int64  
 17  Marital_Married                      10000 non-null  int64  
 18  Marital_Never Married                10000 non-null  int64  
 19  Marital_Separated                    10000 non-null  int64  
 20  Marital_Widowed                      10000 non-null  int64  
 21  Gender_Male                          10000 non-null  int64  
 22  Gender_Nonbinary                     10000 non-null  int64  
 23  Techie_Yes                           10000 non-null  int64  
 24  Contract_One year                    10000 non-null  int64  
 25  Contract_Two Year                    10000 non-null  int64  
 26  Port_modem_Yes                       10000 non-null  int64  
 27  Tablet_Yes                           10000 non-null  int64  
 28  InternetService_Fiber Optic          10000 non-null  int64  
 29  Phone_Yes                            10000 non-null  int64  
 30  Multiple_Yes                         10000 non-null  int64  
 31  OnlineSecurity_Yes                   10000 non-null  int64  
 32  OnlineBackup_Yes                     10000 non-null  int64  
 33  DeviceProtection_Yes                 10000 non-null  int64  
 34  TechSupport_Yes                      10000 non-null  int64  
 35  StreamingTV_Yes                      10000 non-null  int64  
 36  StreamingMovies_Yes                  10000 non-null  int64  
 37  PaperlessBilling_Yes                 10000 non-null  int64  
 38  PaymentMethod_Credit Card (automatic)  10000 non-null  int64  
 39  PaymentMethod_Electronic Check       10000 non-null  int64  
 40  PaymentMethod_Mailed Check           10000 non-null  int64  
dtypes: float64(3), int64(38)
memory usage: 3.1 MB
```

C3: VISUALIZATIONS

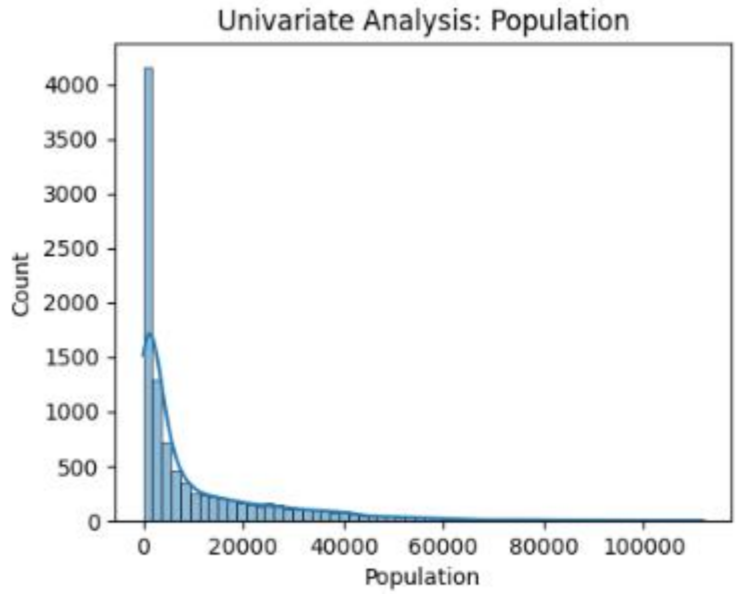
Generate univariate and bivariate visualizations of the distributions of the dependent and independent variables, including the dependent variable in your bivariate visualizations.

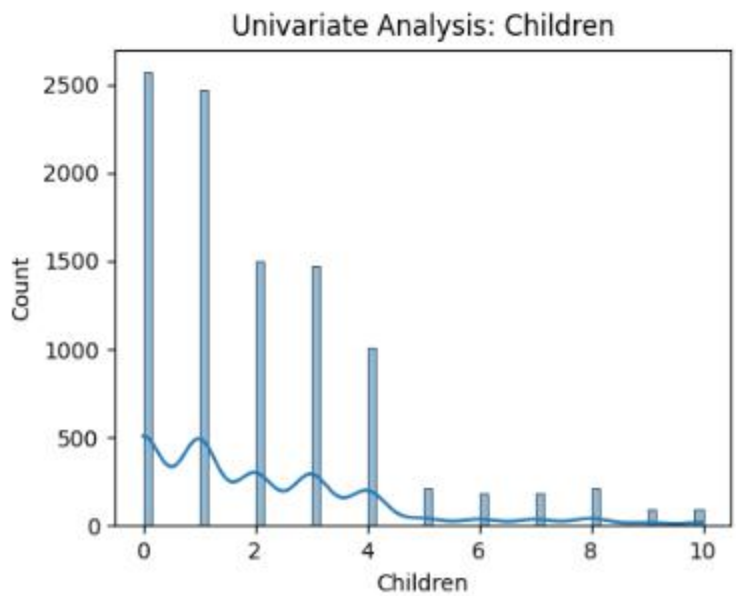
Univariate visualization

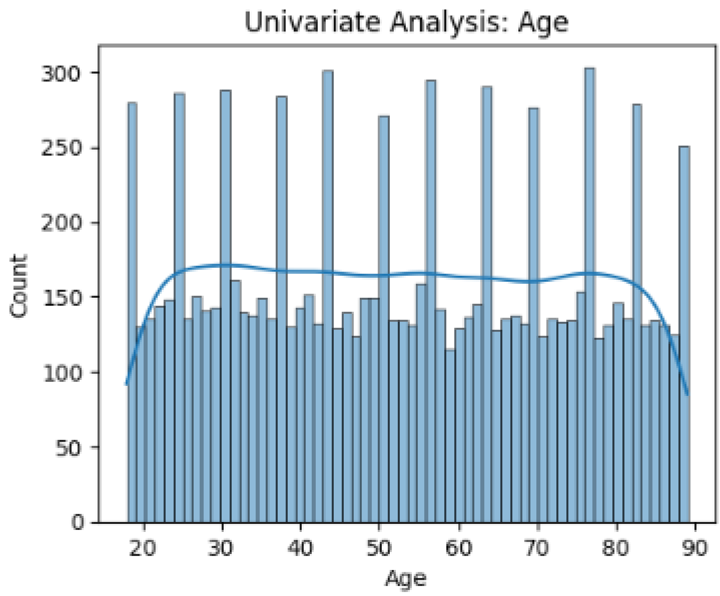
In generating the univariate visualizations, the Monthly charge has been utilized as the dependent variable and all the other variables as independent variables.

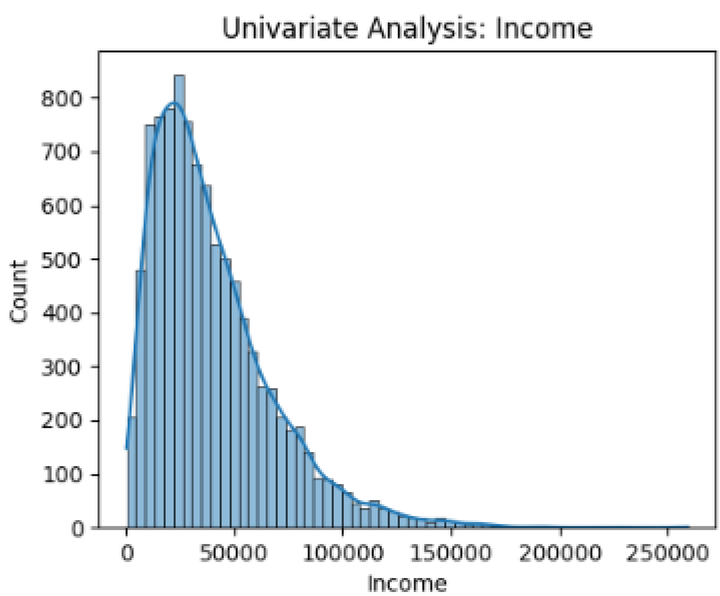


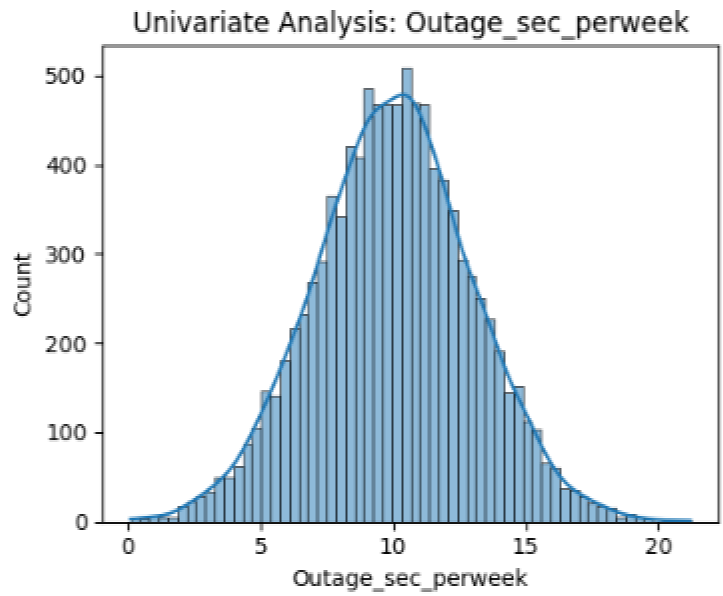
```
# UNIVARIATE ANALYSIS
```

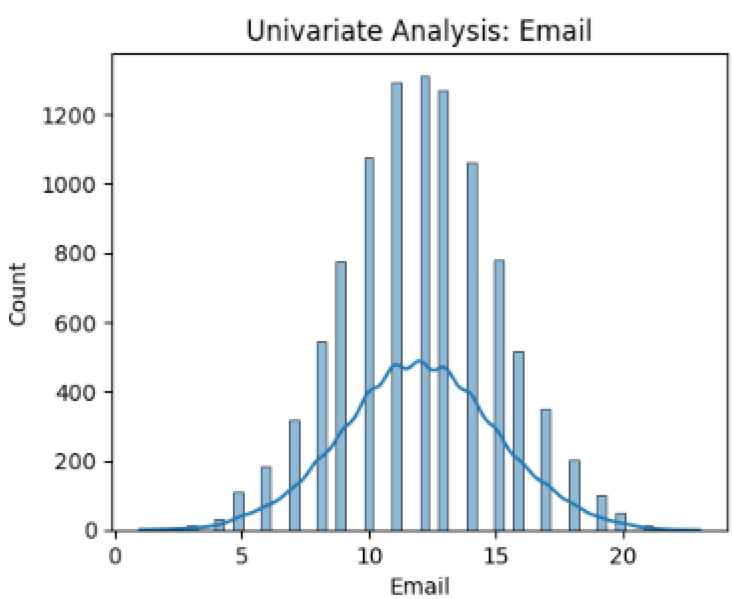


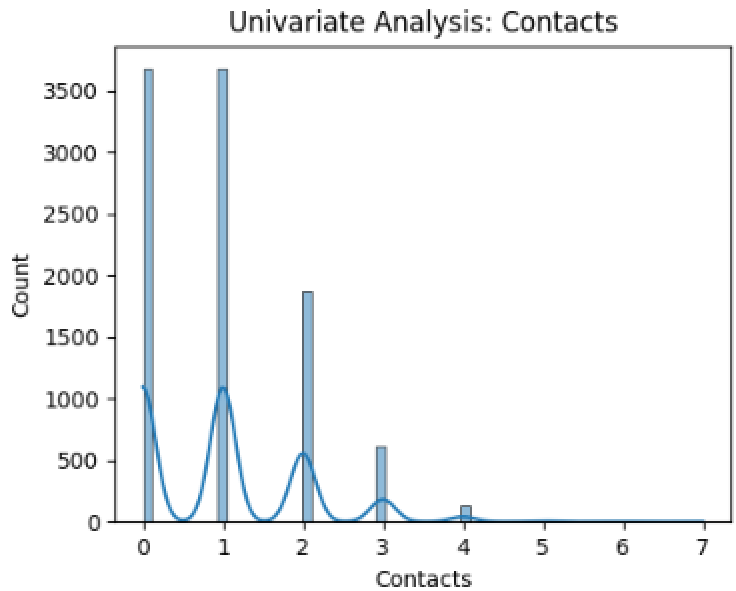


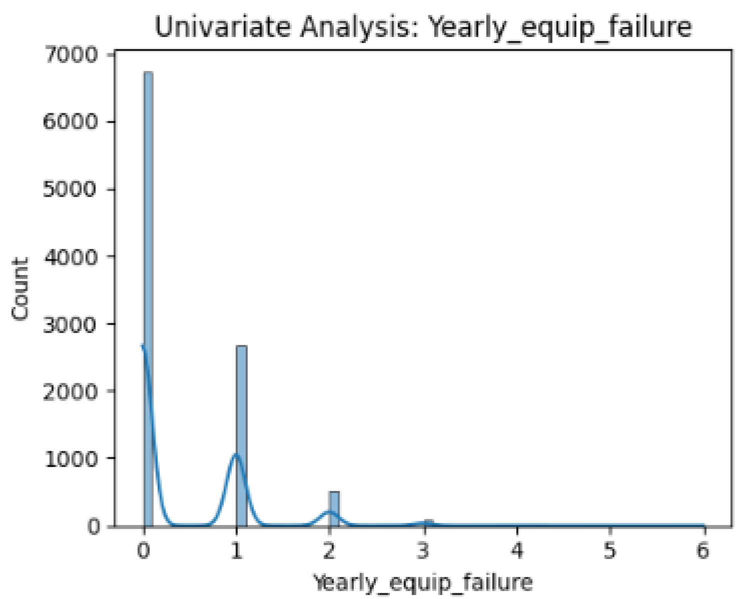

Univariate Analysis: Age
Count
Age
0
50
100
150
200
250
300
20
30
40
50
60
70
80
90

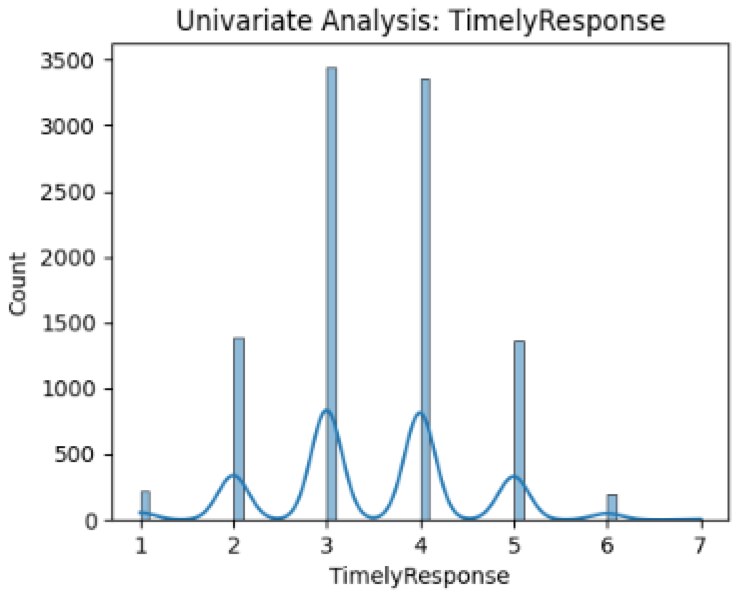

Univariate Analysis: Income
Count
Income
0
100
200
300
400
500
600
700
800
0
50000
100000
150000
200000
250000

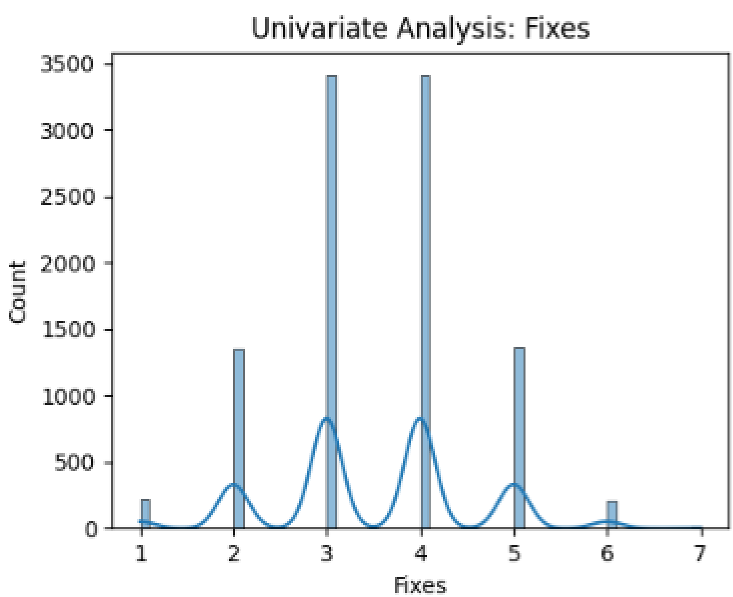

Univariate Analysis: Outage_sec_perweek
Count
Outage_sec_perweek
0
100
200
300
400
500
0
5
10
15
20

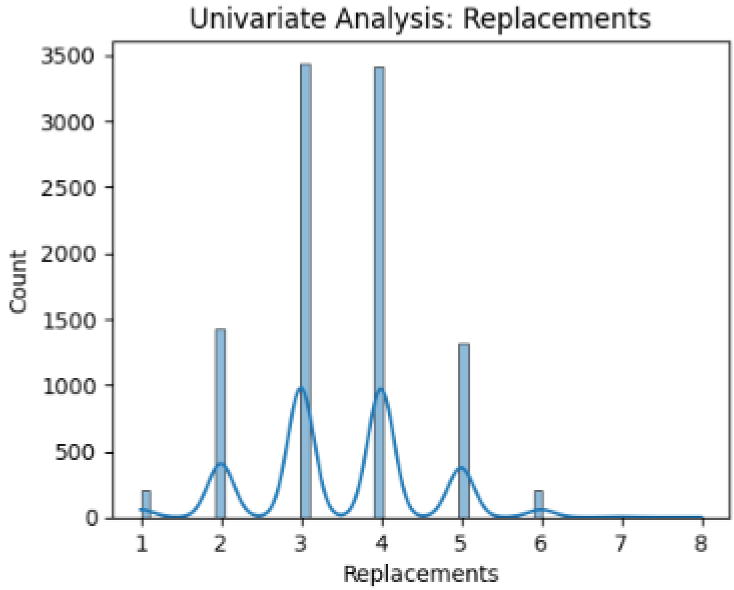

Univariate Analysis: Email
Count
Email
0
200
400
600
800
1000
1200
0
5
10
15
20

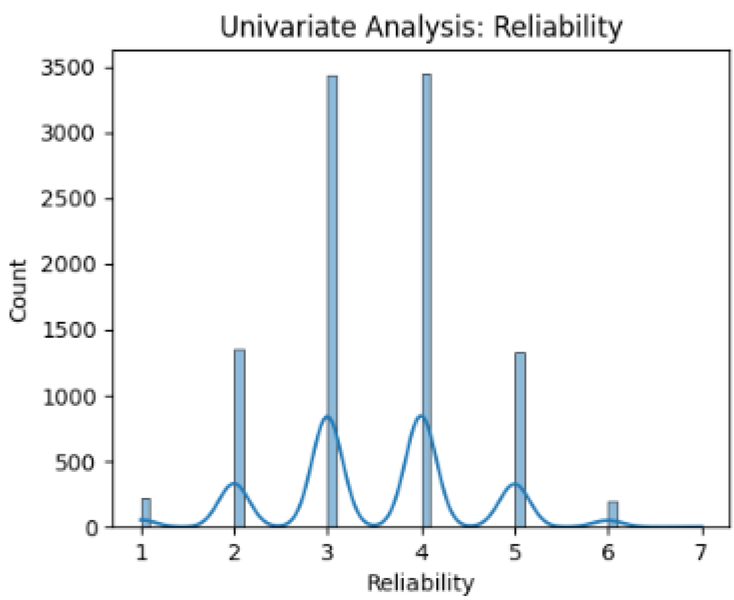

Univariate Analysis: Contacts
Count
Contacts
0
500
1000
1500
2000
2500
3000
3500
0 1 2 3 4 5 6 7

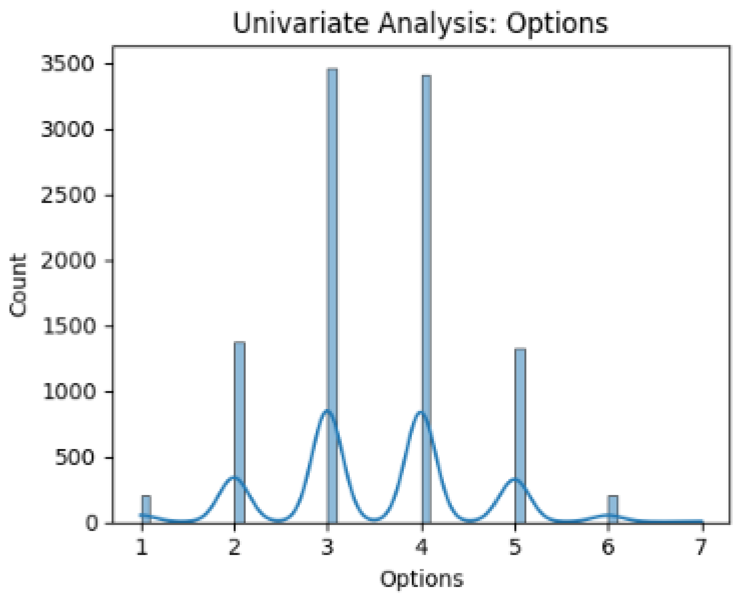

Univariate Analysis: Yearly_equip_failure
Count
Yearly_equip_failure
0
1000
2000
3000
4000
5000
6000
7000
0 1 2 3 4 5 6

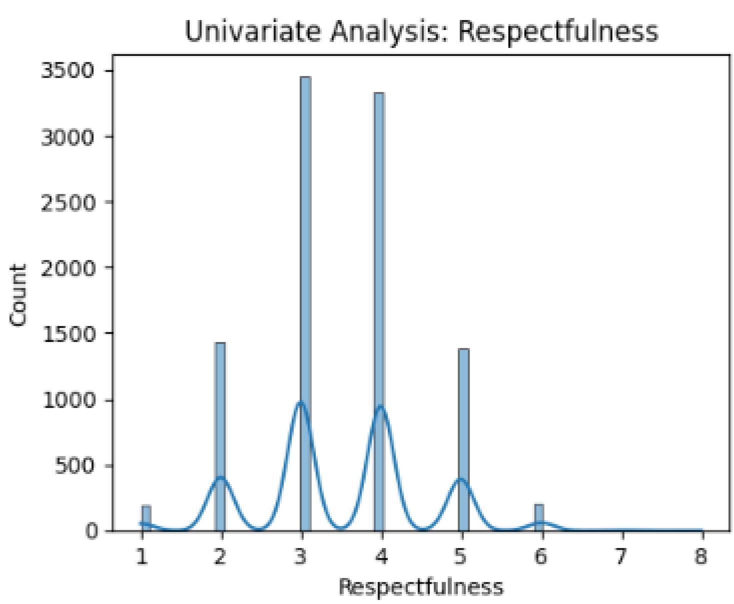

Univariate Analysis: TimelyResponse
Count
TimelyResponse
0
500
1000
1500
2000
2500
3000
3500
1
2
3
4
5
6
7

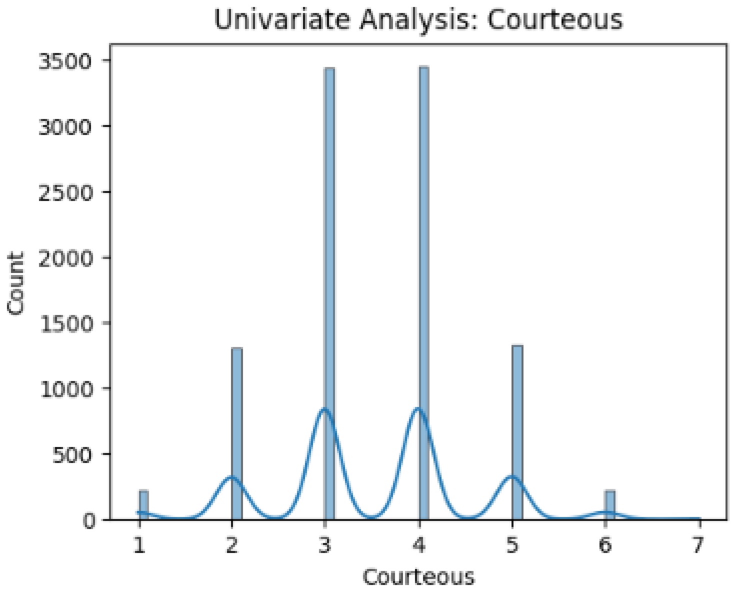

Univariate Analysis: Fixes
Count
Fixes
0
500
1000
1500
2000
2500
3000
3500
1
2
3
4
5
6
7

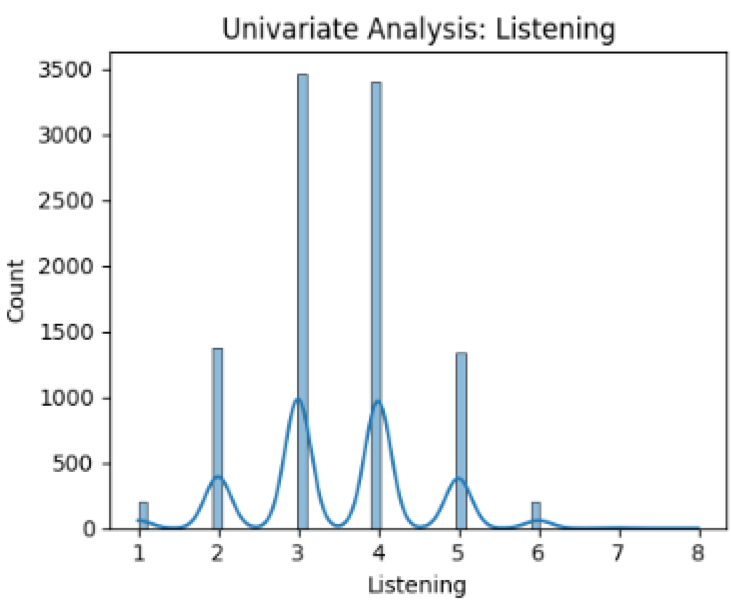

Univariate Analysis: Replacements
Count
Replacements
3500
3000
2500
2000
1500
1000
500
0
1
2
3
4
5
6
7
8

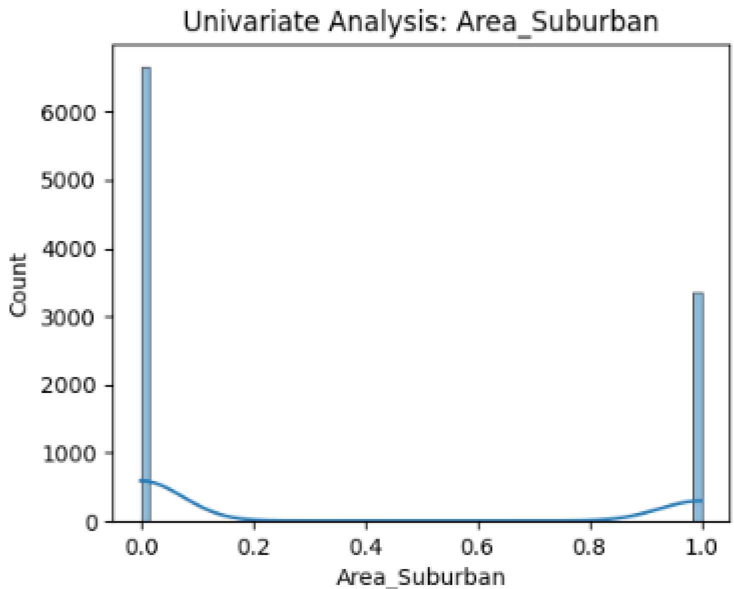

Univariate Analysis: Reliability
Count
Reliability
3500
3000
2500
2000
1500
1000
500
0
1
2
3
4
5
6
7

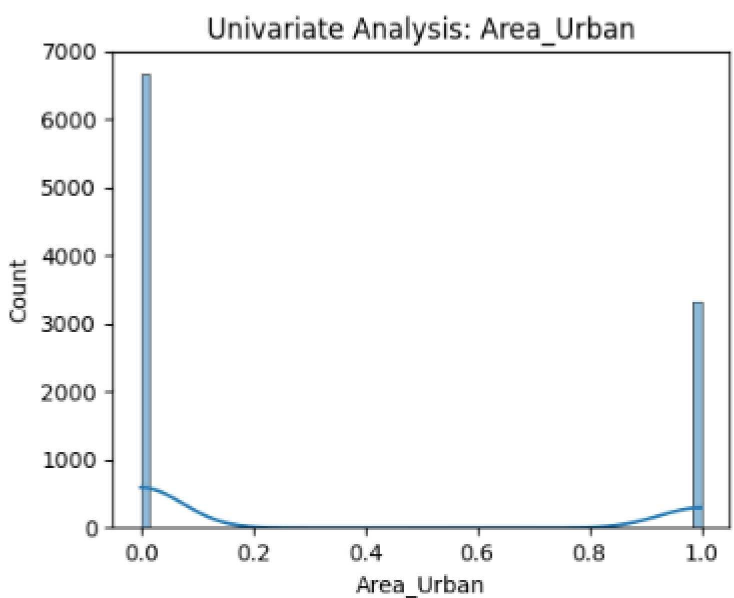

Univariate Analysis: Options
Count
3500
3000
2500
2000
1500
1000
500
0
1
2
3
4
5
6
7
Options

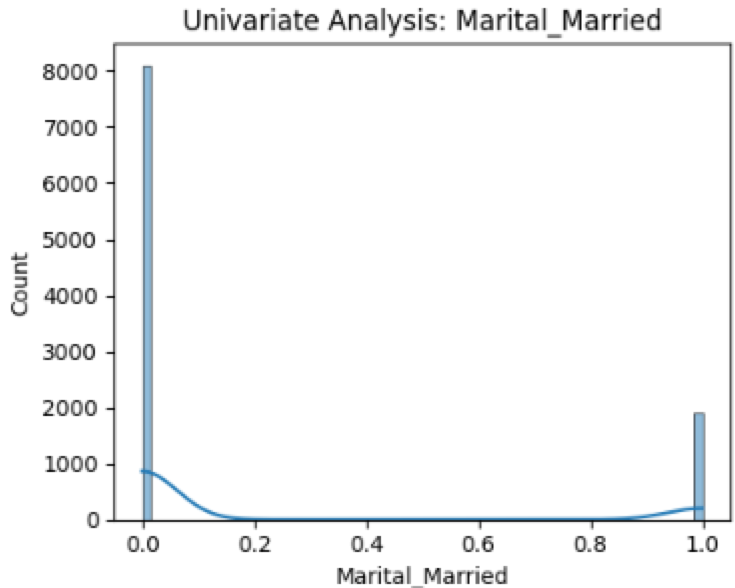

Univariate Analysis: Respectfulness
Count
3500
3000
2500
2000
1500
1000
500
0
1
2
3
4
5
6
7
8
Respectfulness

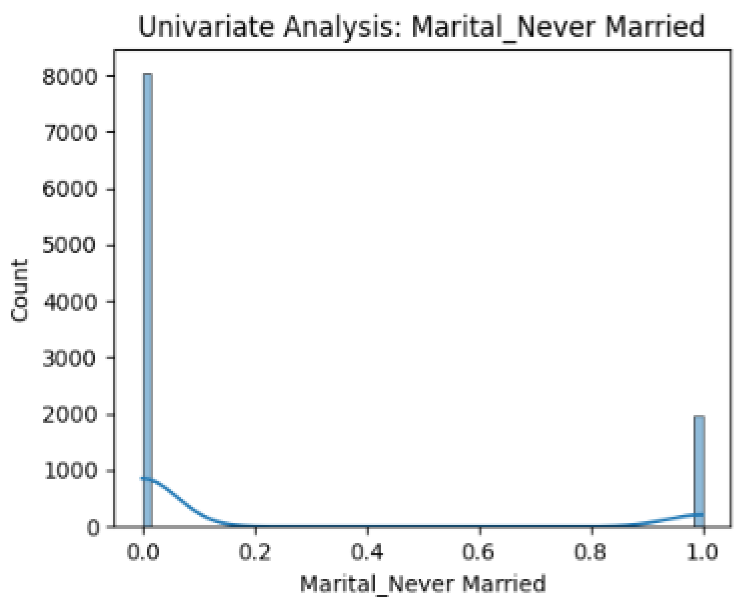

Univariate Analysis: Courteous
Count
Courteous
0
500
1000
1500
2000
2500
3000
3500
1
2
3
4
5
6
7

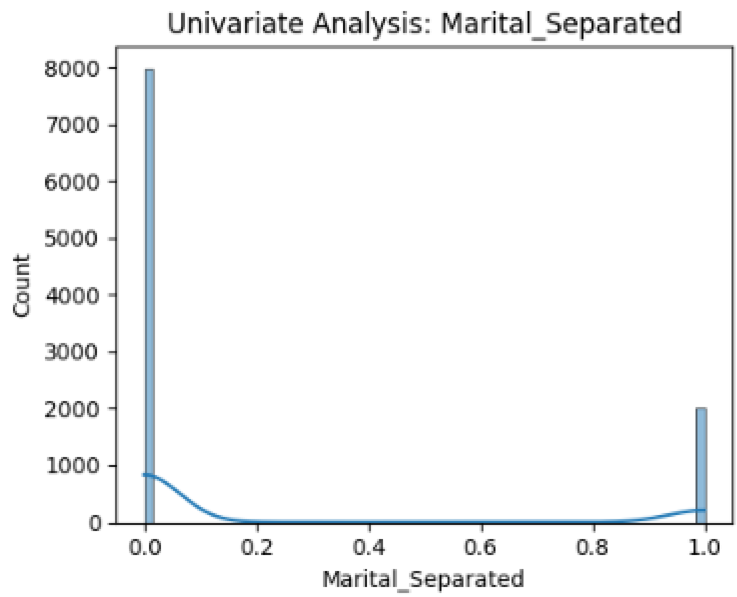

Univariate Analysis: Listening
Count
Listening
0
500
1000
1500
2000
2500
3000
3500
1
2
3
4
5
6
7
8

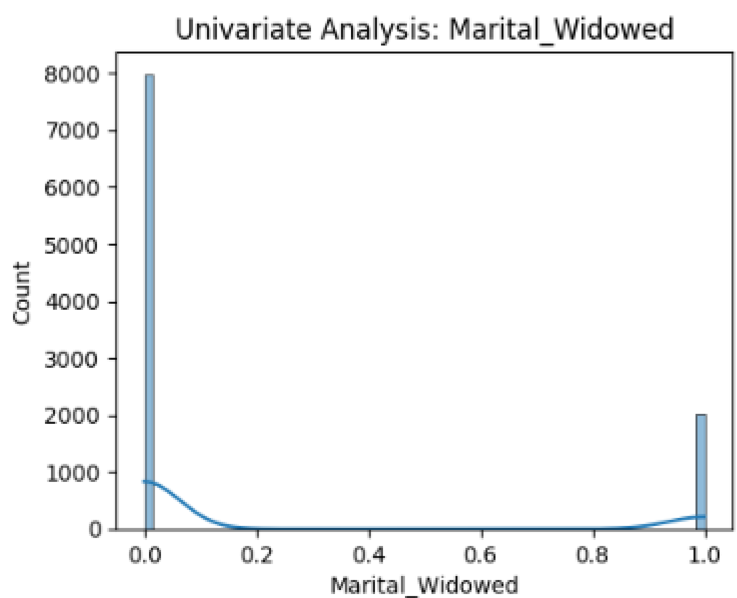

Univariate Analysis: Area_Suburban
Count
6000
5000
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
Area_Suburban

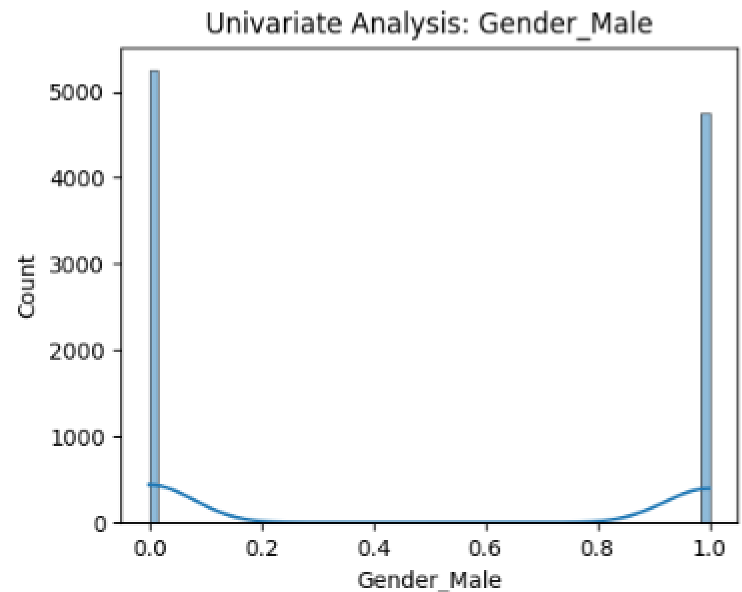

Univariate Analysis: Area_Urban
Count
7000
6000
5000
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
Area_Urban

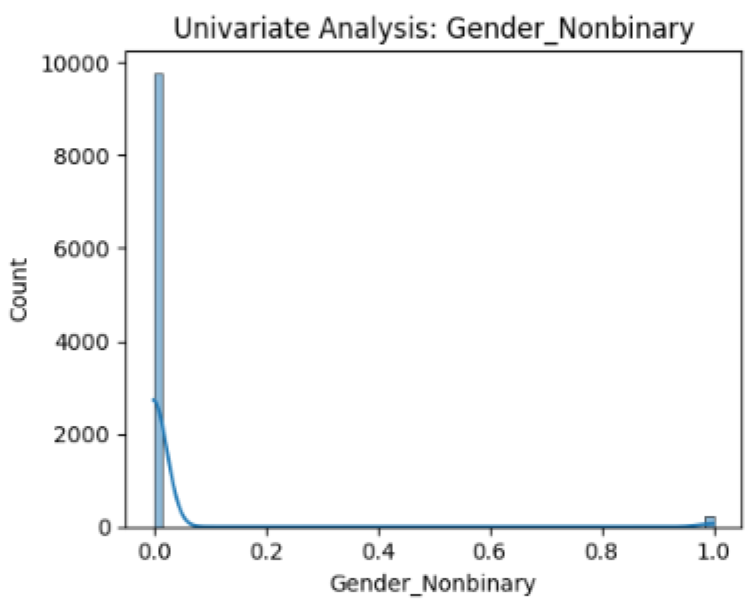

Univariate Analysis: Marital_Married
Count
Marital_Married
0.0
0.2
0.4
0.6
0.8
1.0
0
1000
2000
3000
4000
5000
6000
7000
8000

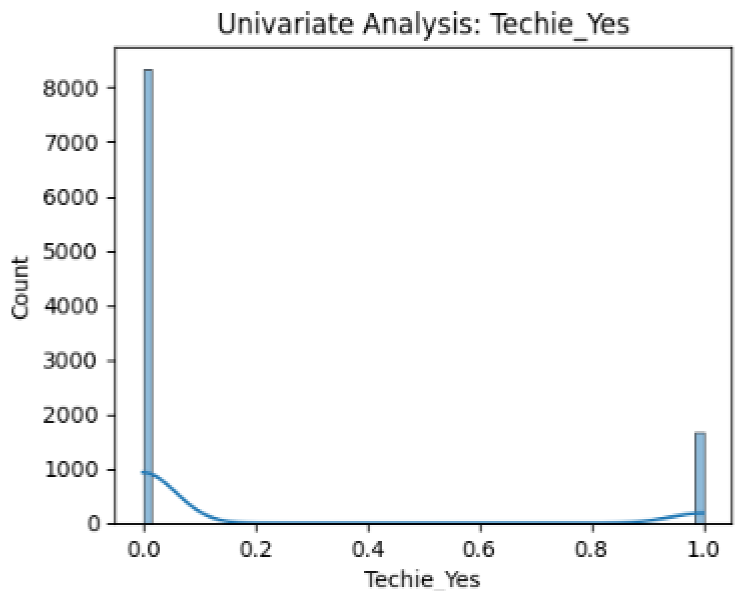

Univariate Analysis: Marital_Never Married
Count
Marital_Never Married
0.0
0.2
0.4
0.6
0.8
1.0
0
1000
2000
3000
4000
5000
6000
7000
8000

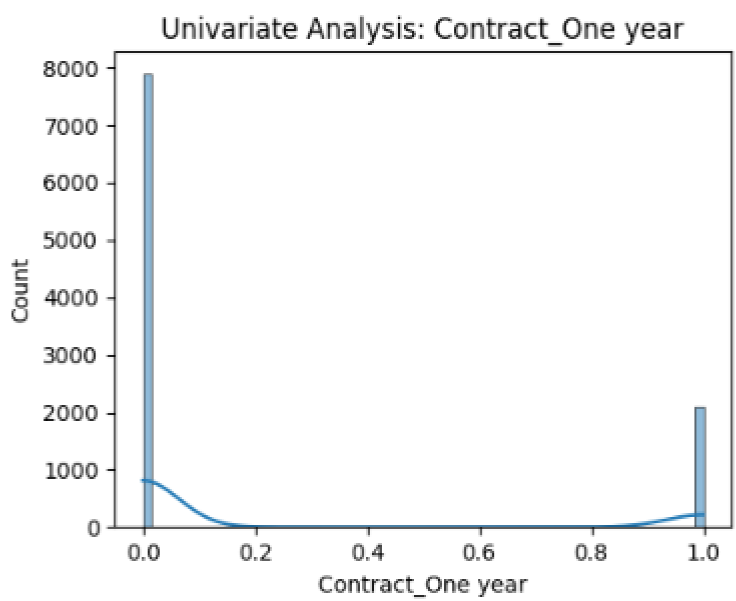

Univariate Analysis: Marital_Separated
8000
7000
6000
5000
4000
3000
2000
1000
0
Count
0.0
0.2
0.4
0.6
0.8
1.0
Marital_Separated

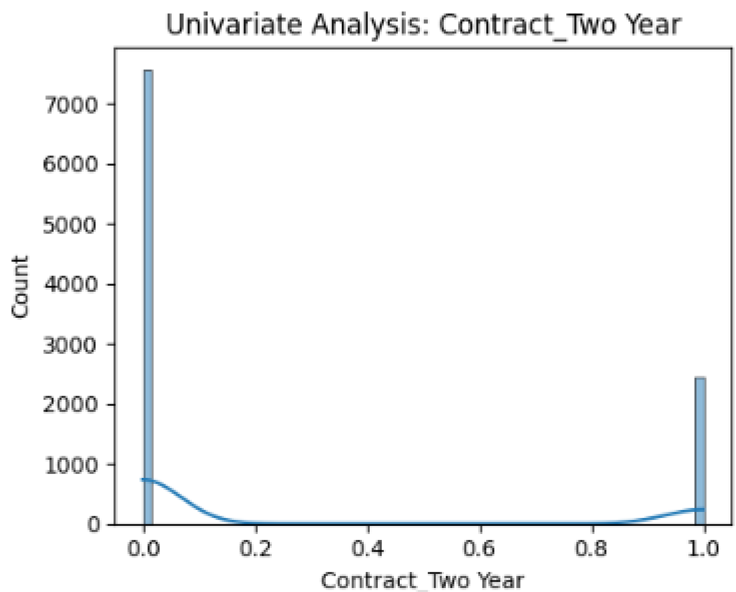

Univariate Analysis: Marital_Widowed
8000
7000
6000
5000
4000
3000
2000
1000
0
Count
0.0
0.2
0.4
0.6
0.8
1.0
Marital_Widowed

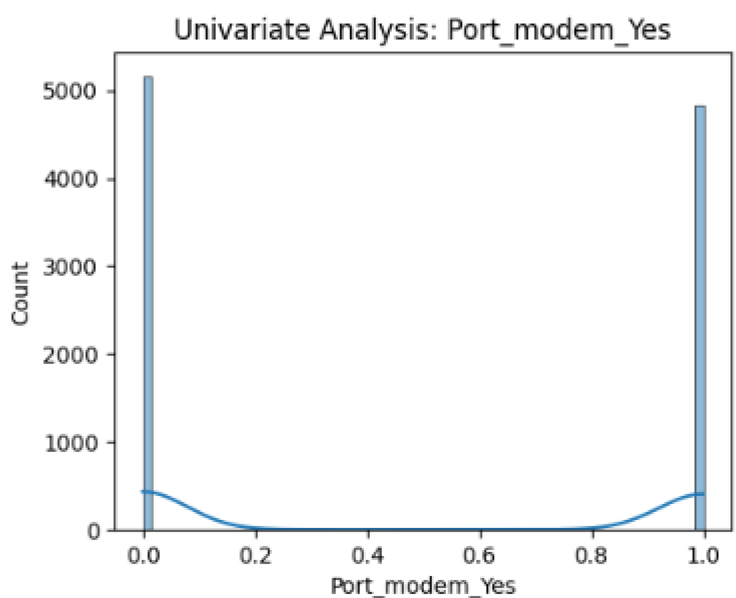

Univariate Analysis: Gender_Male
Count
Gender_Male
0
1000
2000
3000
4000
5000
0.0
0.2
0.4
0.6
0.8
1.0

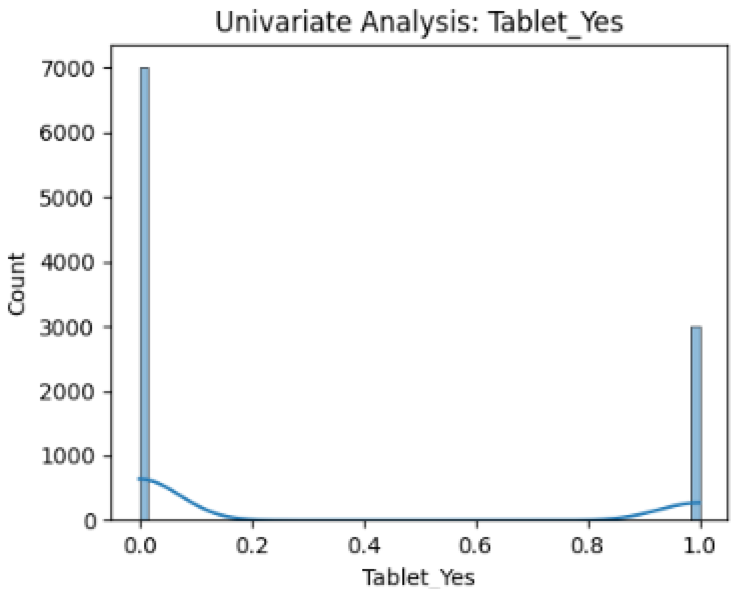

Univariate Analysis: Gender_Nonbinary
Count
Gender_Nonbinary
0
2000
4000
6000
8000
10000
0.0
0.2
0.4
0.6
0.8
1.0

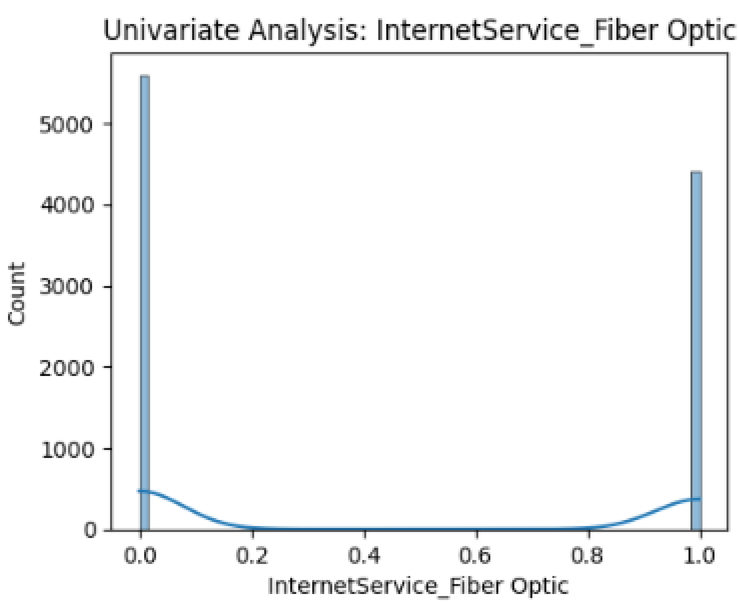

Univariate Analysis: Techie_Yes
Count
Techie_Yes
0
1000
2000
3000
4000
5000
6000
7000
8000
0.0
0.2
0.4
0.6
0.8
1.0

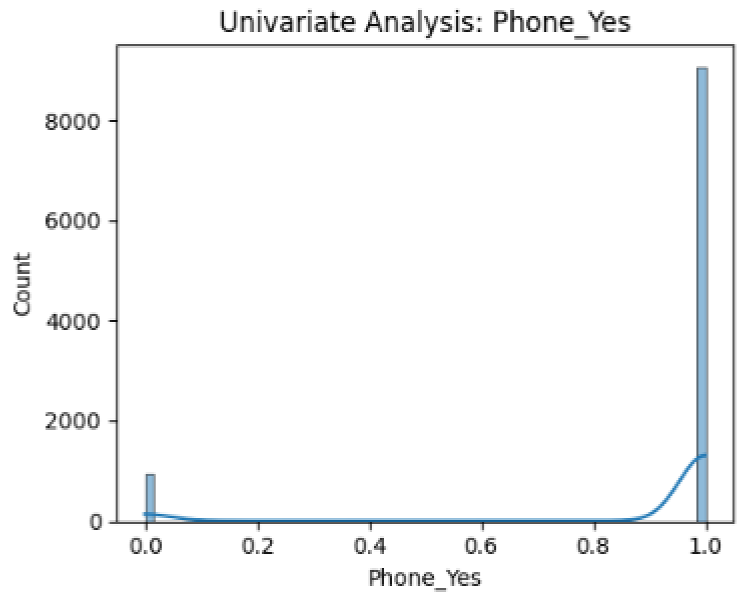

Univariate Analysis: Contract_One year
Count
Contract_One year
0
1000
2000
3000
4000
5000
6000
7000
8000
0.0
0.2
0.4
0.6
0.8
1.0

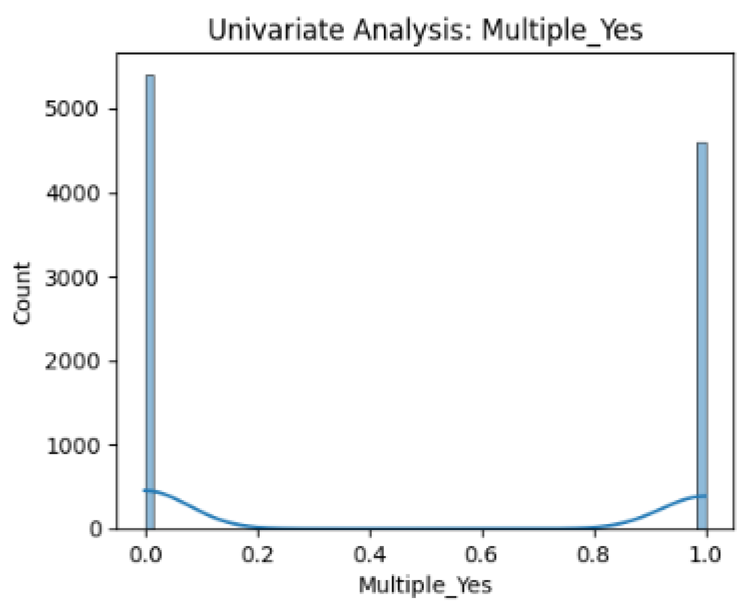

Univariate Analysis: Contract_Two Year
Count
7000
6000
5000
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
Contract_Two Year

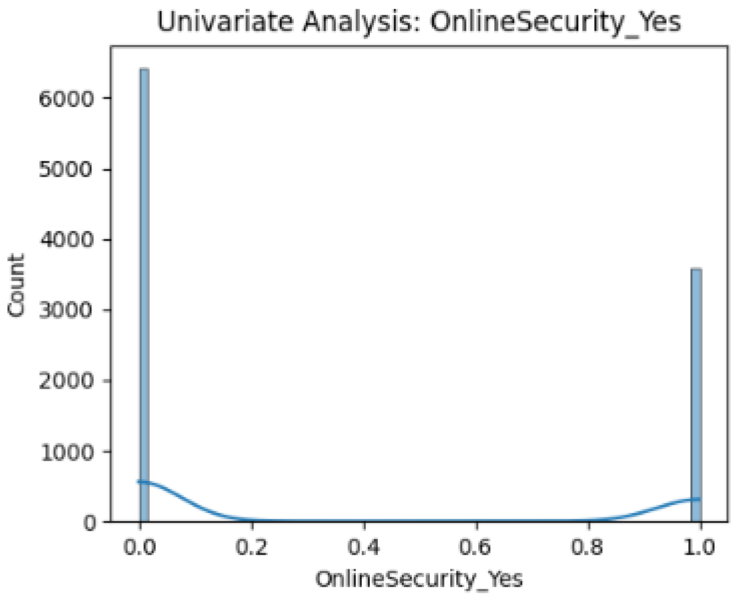

Univariate Analysis: Port_modem_Yes
Count
5000
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
Port_modem_Yes

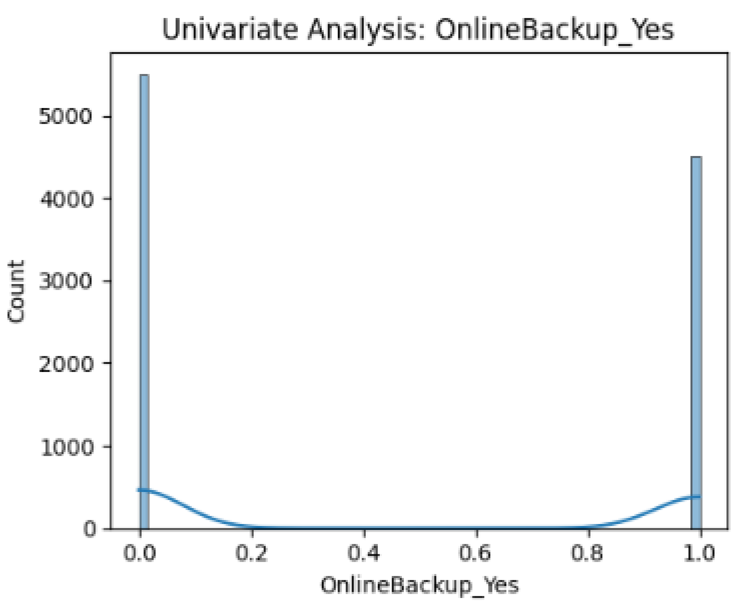

Univariate Analysis: Tablet_Yes
Count
7000
6000
5000
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
Tablet_Yes

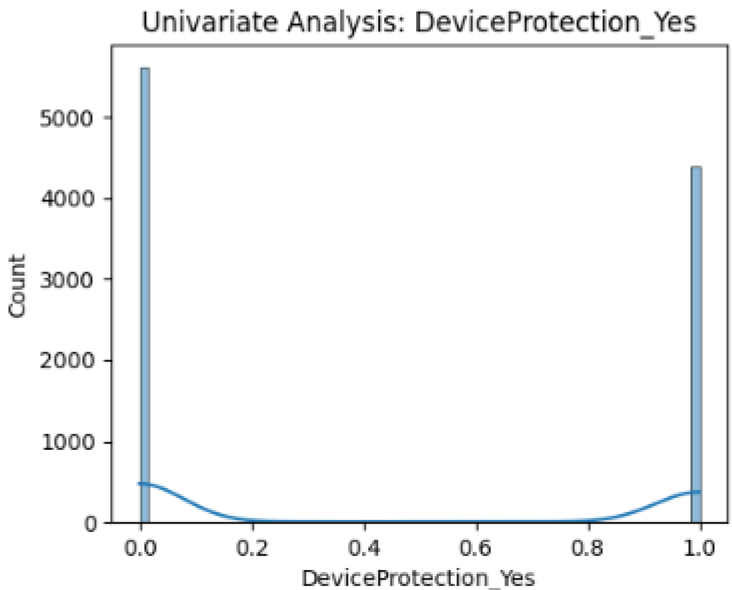

Univariate Analysis: InternetService_Fiber Optic
Count
5000
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
InternetService_Fiber Optic

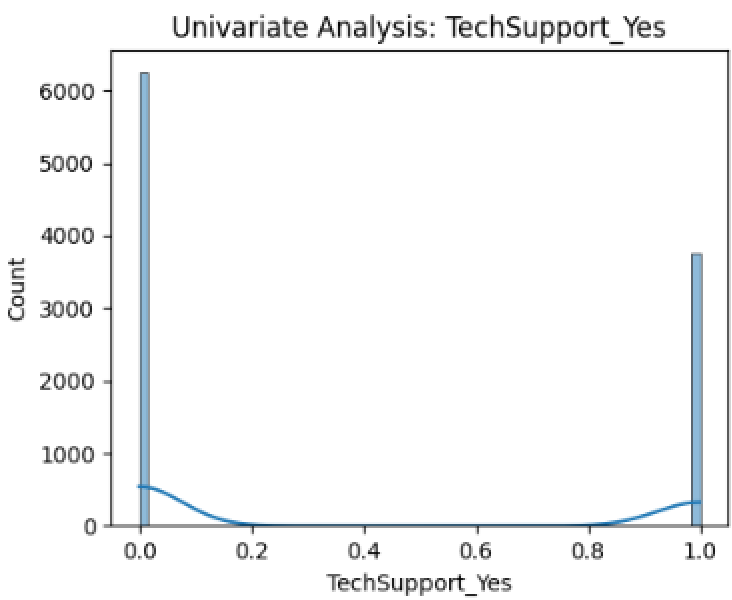

Univariate Analysis: Phone_Yes
Count
8000
6000
4000
2000
0
0.0
0.2
0.4
0.6
0.8
1.0
Phone_Yes

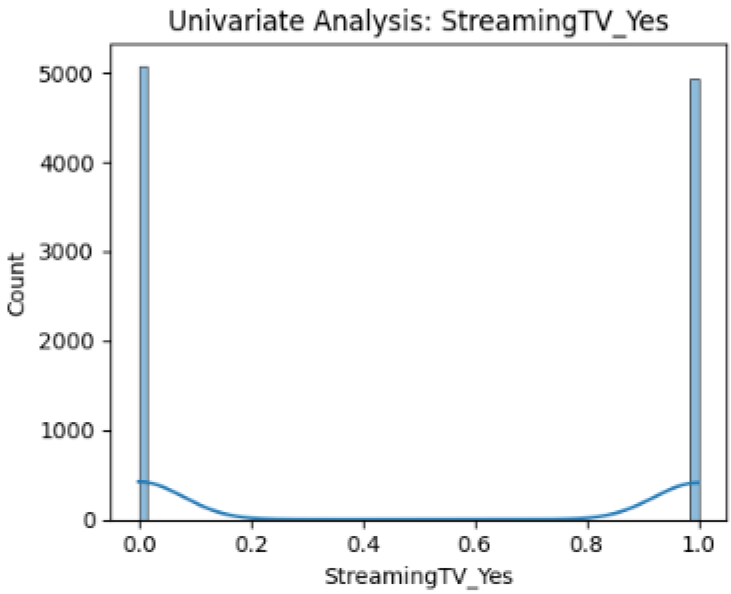

Univariate Analysis: Multiple_Yes
Count
5000
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
Multiple_Yes

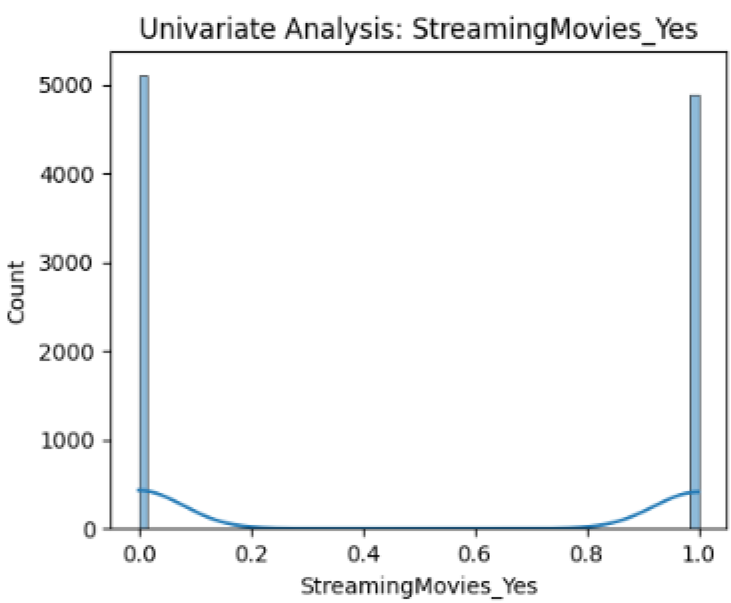

Univariate Analysis: OnlineSecurity_Yes
Count
6000
5000
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
OnlineSecurity_Yes

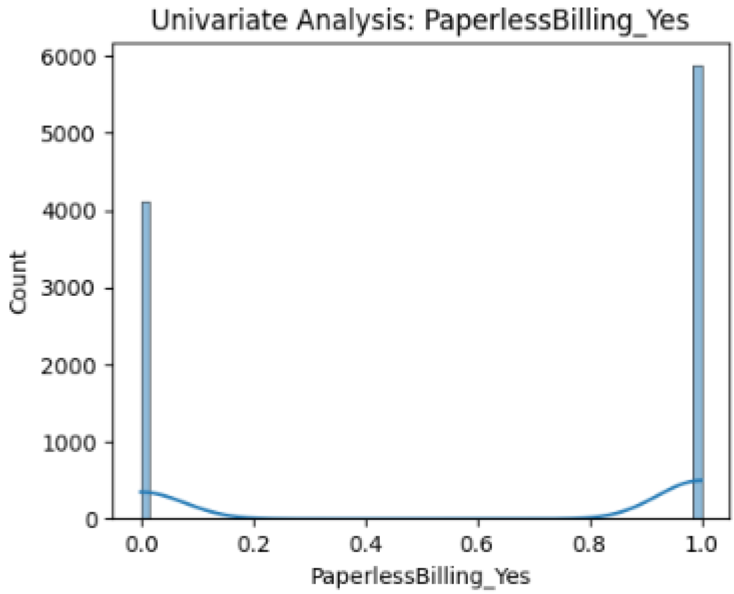

Univariate Analysis: OnlineBackup_Yes
Count
5000
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
OnlineBackup_Yes

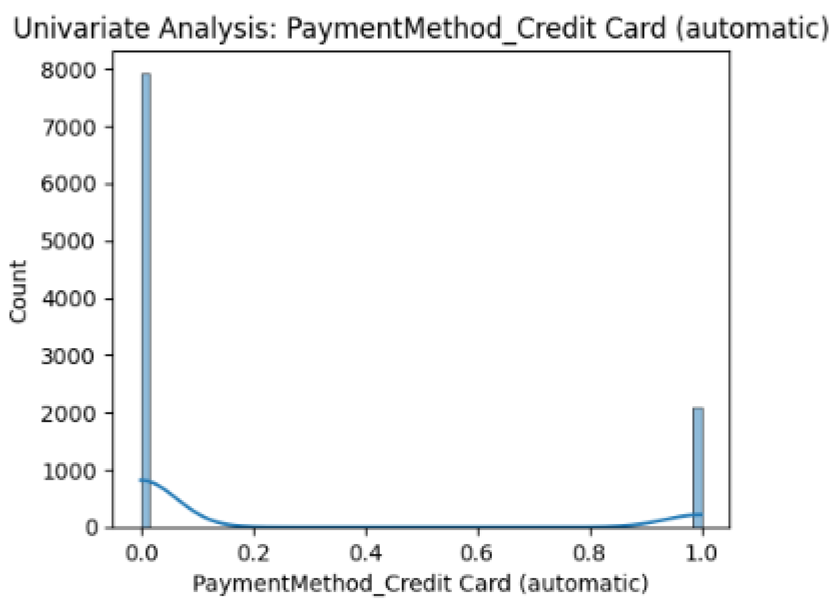

Univariate Analysis: DeviceProtection_Yes
Count
DeviceProtection_Yes
0
1000
2000
3000
4000
5000
0.0
0.2
0.4
0.6
0.8
1.0

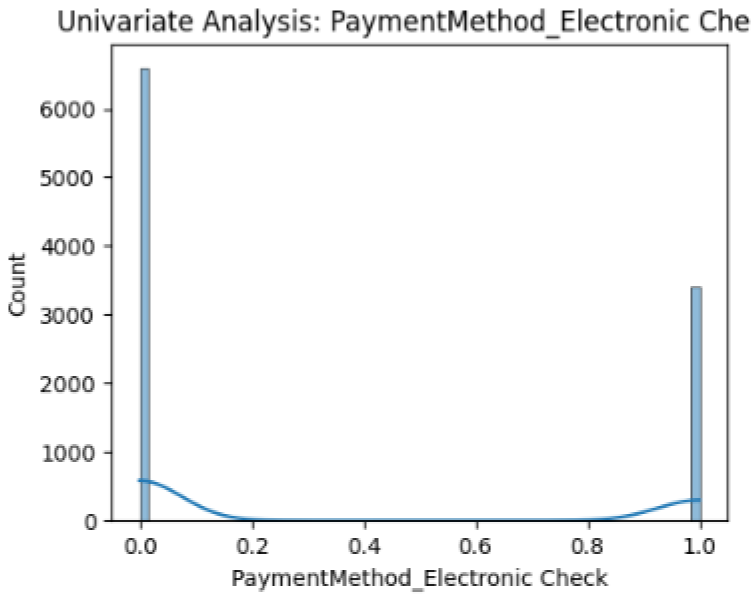

Univariate Analysis: TechSupport_Yes
Count
TechSupport_Yes
0
1000
2000
3000
4000
5000
6000
0.0
0.2
0.4
0.6
0.8
1.0

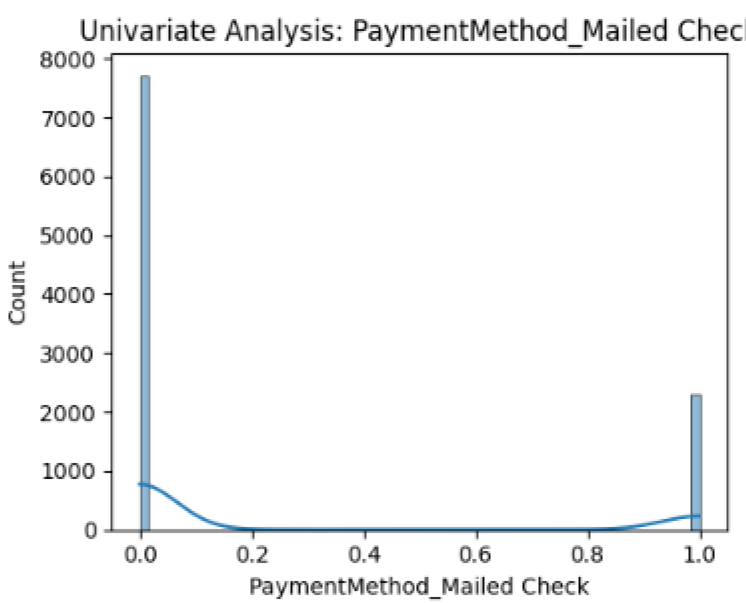

Univariate Analysis: StreamingTV_Yes
Count
5000
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
StreamingTV_Yes


Univariate Analysis: StreamingMovies_Yes
Count
5000
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
StreamingMovies_Yes


Univariate Analysis: PaperlessBilling_Yes
6000
5000
4000
3000
2000
1000
0
Count
0.0
0.2
0.4
0.6
0.8
1.0
PaperlessBilling_Yes

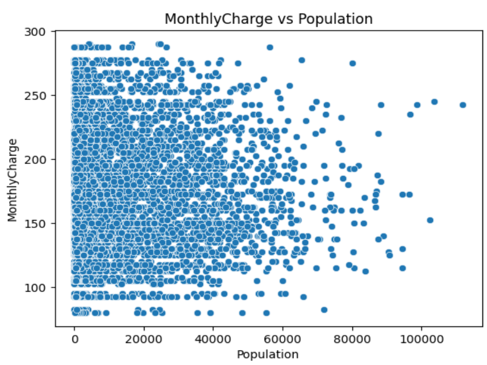

Univariate Analysis: PaymentMethod_Credit Card (automatic)
8000
7000
6000
5000
4000
3000
2000
1000
0
Count
0.0
0.2
0.4
0.6
0.8
1.0
PaymentMethod_Credit Card (automatic)

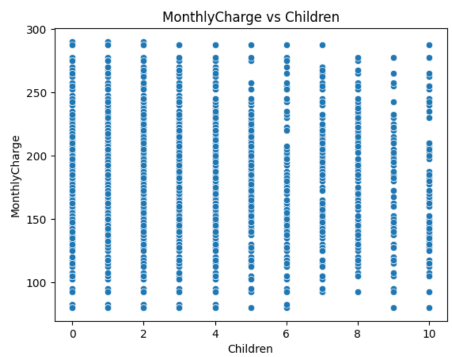

Univariate Analysis: PaymentMethod_Electronic Che
Count
0
1000
2000
3000
4000
5000
6000
0.0
0.2
0.4
0.6
0.8
1.0
PaymentMethod_Electronic Check

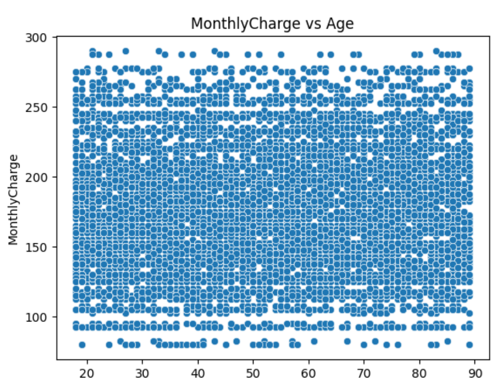

Univariate Analysis: PaymentMethod_Mailed Checl
Count
0
1000
2000
3000
4000
5000
6000
7000
8000
0.0
0.2
0.4
0.6
0.8
1.0
PaymentMethod_Mailed Check

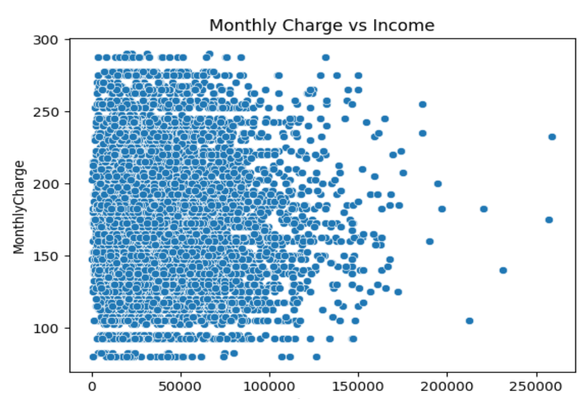
Bivariate visualization

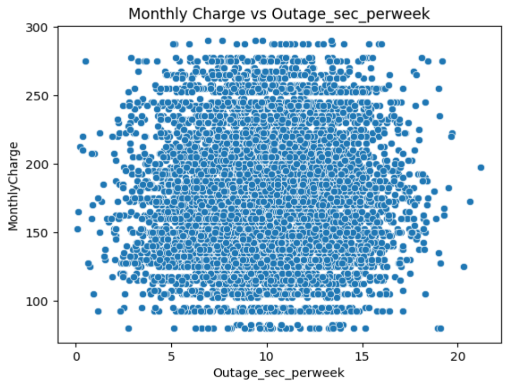
In the generation of the bivariate visualizations, the Monthly charge has been utilized as the dependent variable and has been plotted against the independent variables

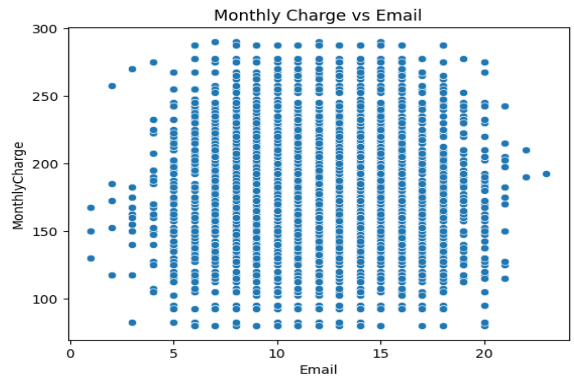


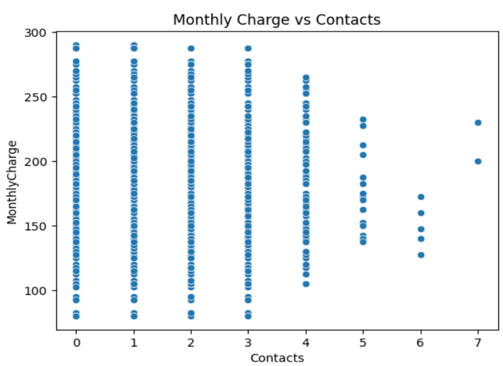


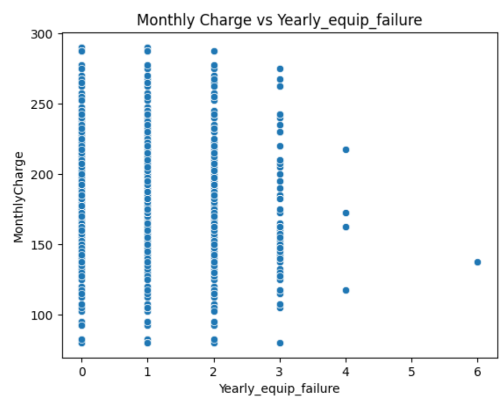


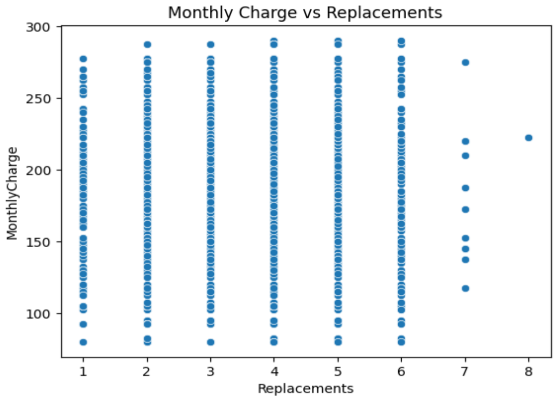


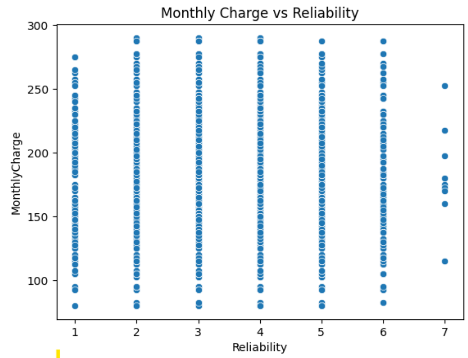

Monthly Charge vs Outage_sec_perweek
MonthlyCharge
Outage_sec_perweek

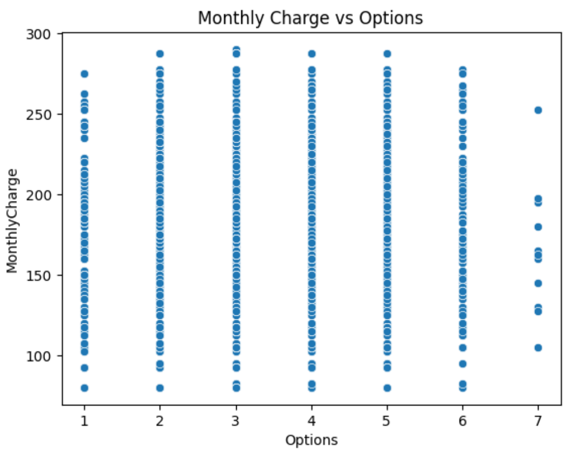

Monthly Charge vs Email
MonthlyCharge
Email

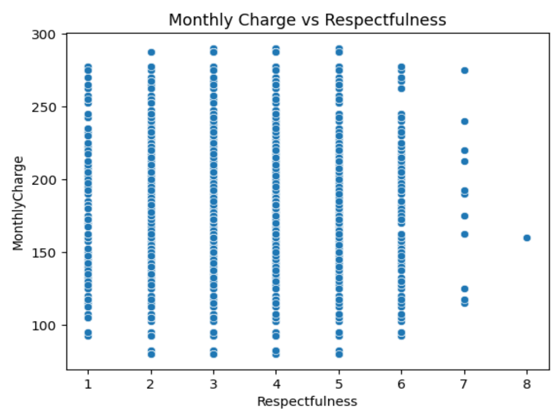

Monthly Charge vs Contacts
MonthlyCharge
Contacts

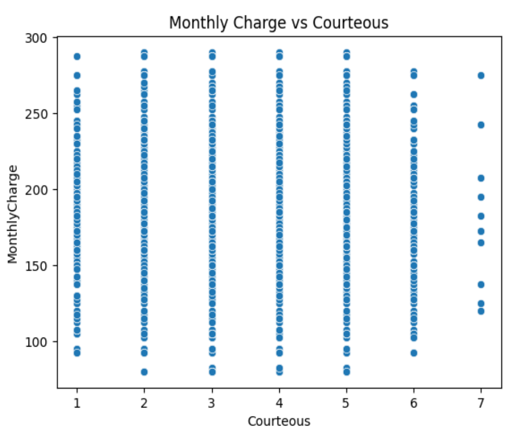

Monthly Charge vs Yearly_equip_failure
MonthlyCharge
Yearly_equip_failure

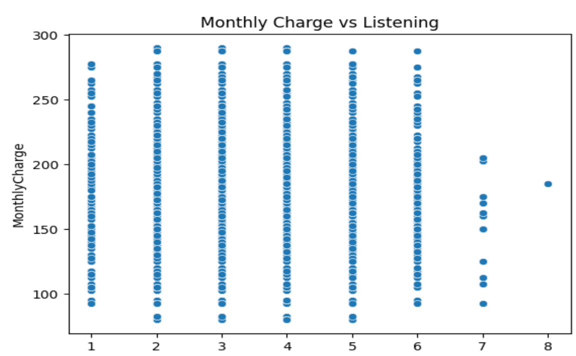

Monthly Charge vs Replacements
MonthlyCharge
300
250
200
150
100
1 2 3 4 5 6 7 8
Replacements

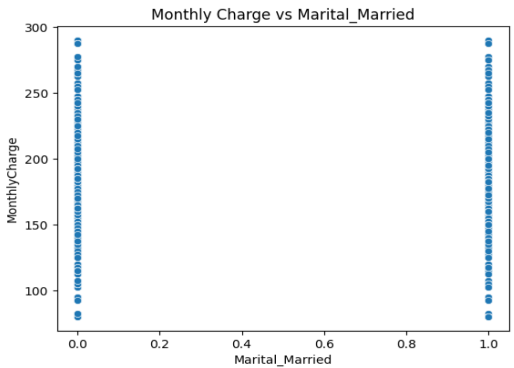

Monthly Charge vs Reliability
MonthlyCharge
300
250
200
150
100
1 2 3 4 5 6 7
Reliability

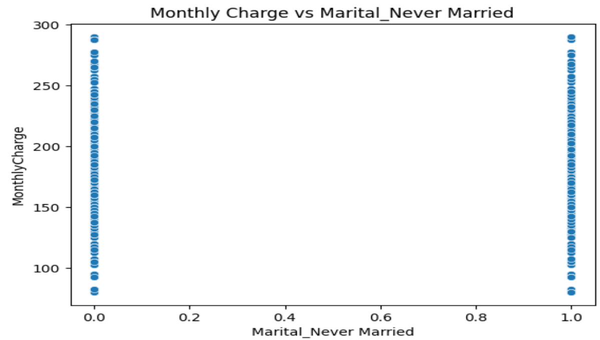

Monthly Charge vs Options
MonthlyCharge
300
250
200
150
100
1 2 3 4 5 6 7
Options

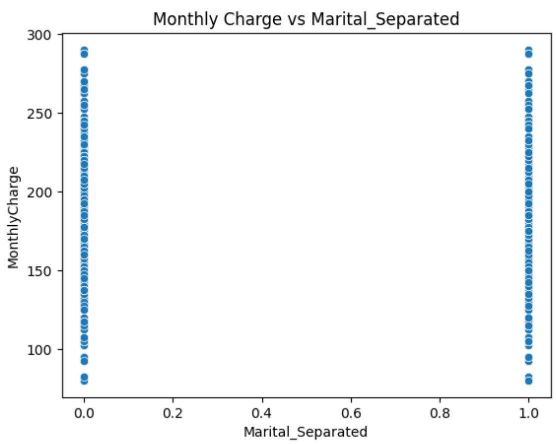

Monthly Charge vs Respectfulness
MonthlyCharge
300
250
200
150
100
1 2 3 4 5 6 7 8
Respectfulness

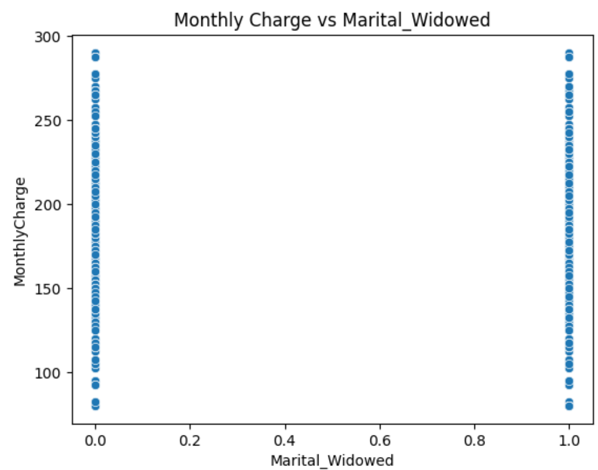

Monthly Charge vs Courteous
MonthlyCharge
300
250
200
150
100
1 2 3 4 5 6 7
Courteous

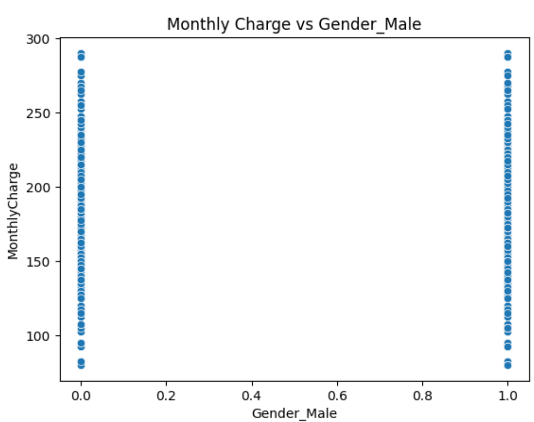

Monthly Charge vs Listening
MonthlyCharge
300
250
200
150
100
1 2 3 4 5 6 7 8
Listening

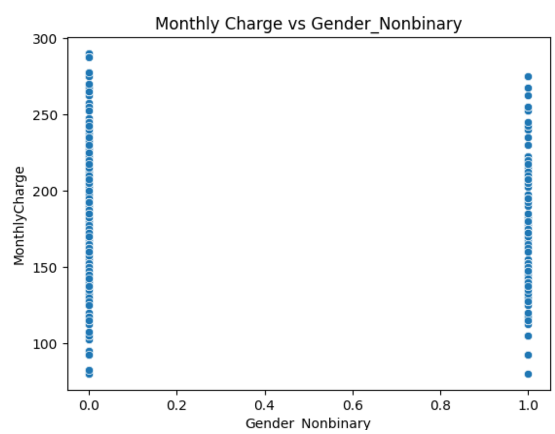

Monthly Charge vs Marital_Married
MonthlyCharge
Marital_Married

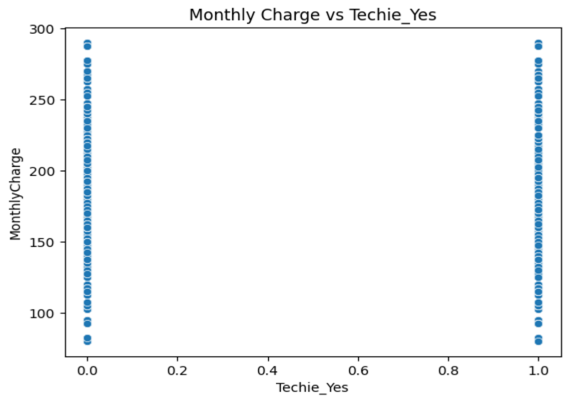

Monthly Charge vs Marital_Never Married
MonthlyCharge
Marital_Never Married

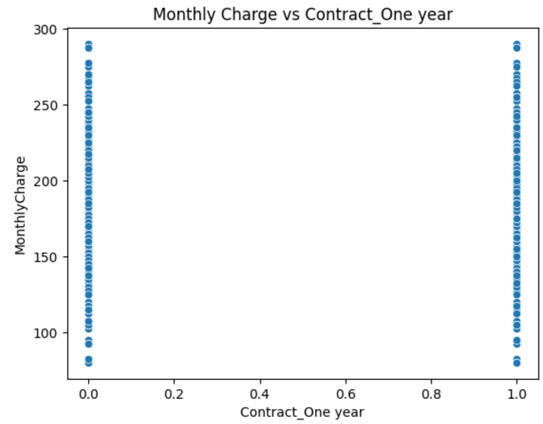

Monthly Charge vs Marital_Separated
MonthlyCharge
Marital_Separated

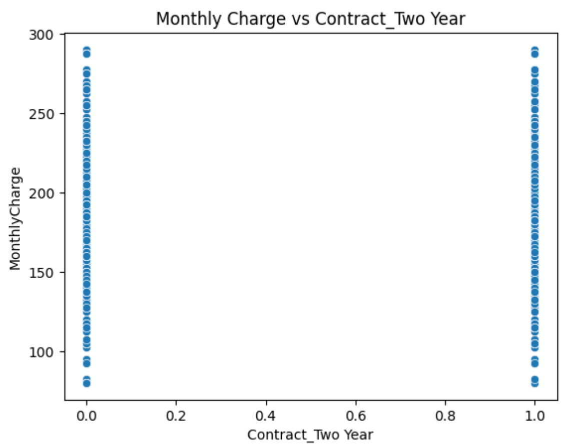

Monthly Charge vs Marital_Widowed
MonthlyCharge
Marital_Widowed

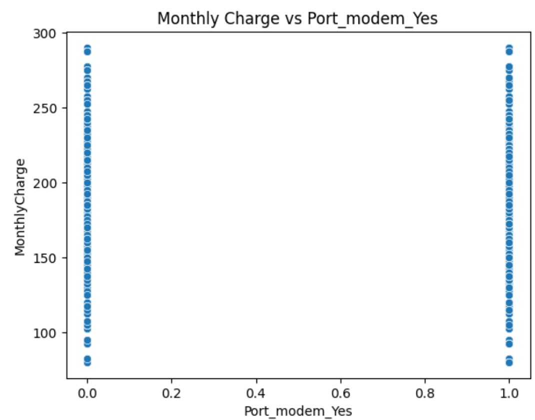

Monthly Charge vs Gender_Male
MonthlyCharge
Gender_Male

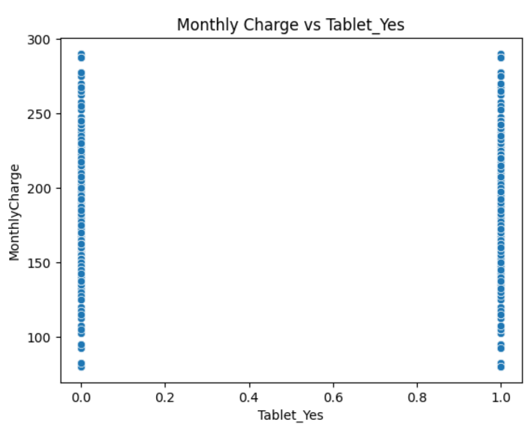

Monthly Charge vs Gender_Nonbinary
MonthlyCharge
Gender_Nonbinary

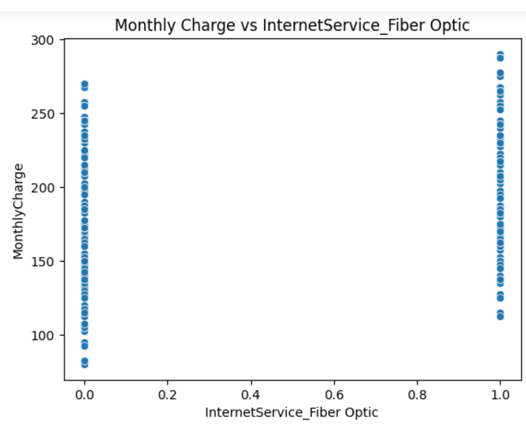

Monthly Charge vs Techie_Yes
MonthlyCharge
Techie_Yes

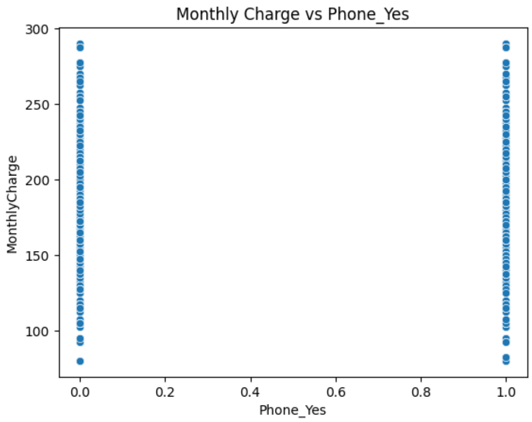

Monthly Charge vs Contract_One year
MonthlyCharge
Contract_One year

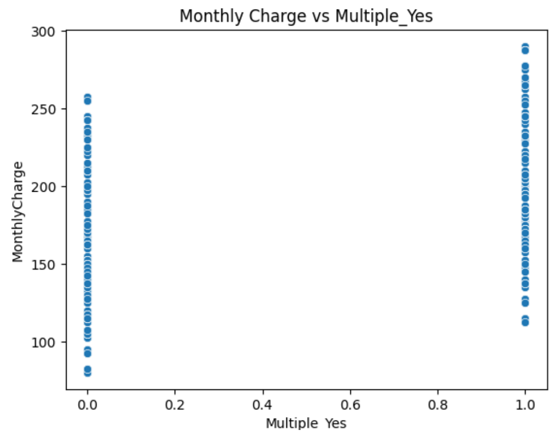

Monthly Charge vs Contract_Two Year
MonthlyCharge
Contract_Two Year

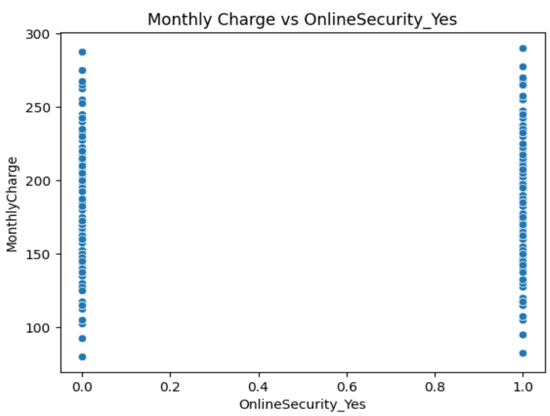

Monthly Charge vs Port_modem_Yes
MonthlyCharge
Port_modem_Yes

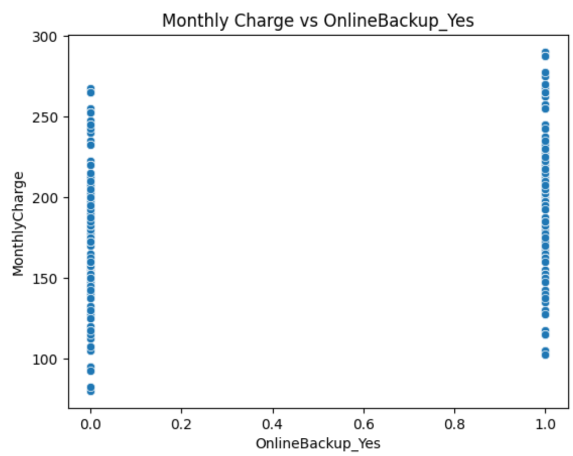

Monthly Charge vs Tablet_Yes
MonthlyCharge
Tablet_Yes

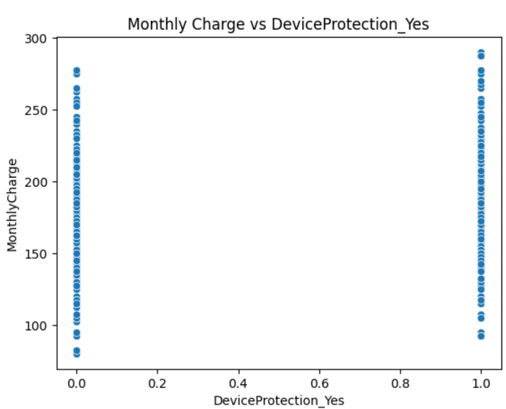

Monthly Charge vs InternetService_Fiber Optic
MonthlyCharge
InternetService_Fiber Optic

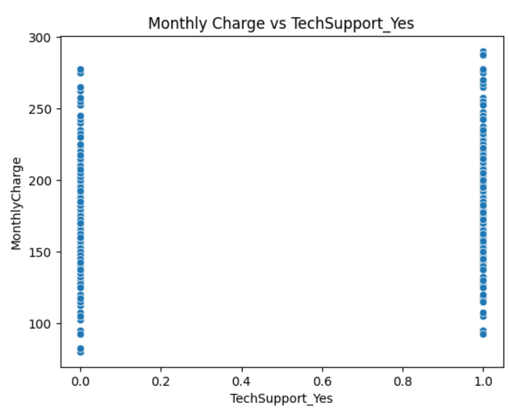

Monthly Charge vs Phone_Yes
MonthlyCharge
Phone_Yes

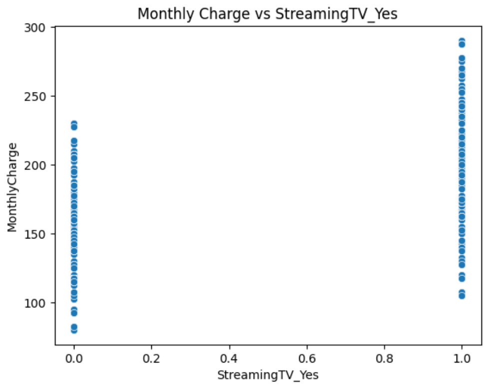

Monthly Charge vs Multiple_Yes
MonthlyCharge
Multiple_Yes

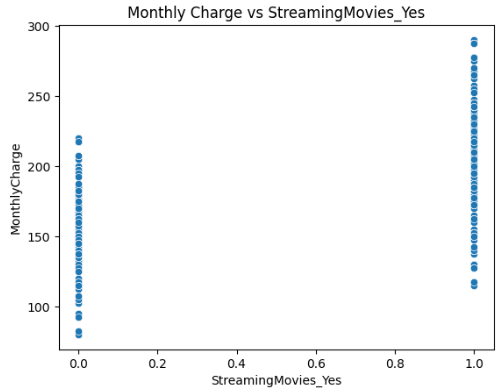

Monthly Charge vs OnlineSecurity_Yes
MonthlyCharge
OnlineSecurity_Yes

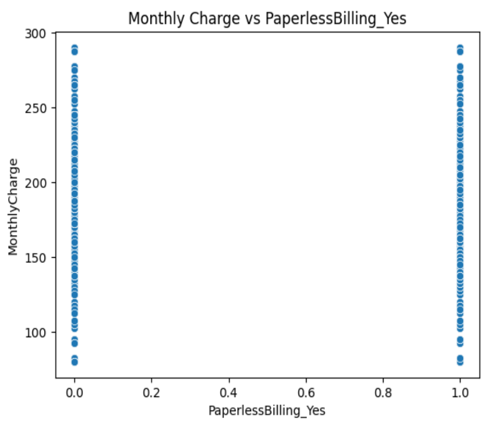

Monthly Charge vs OnlineBackup_Yes
MonthlyCharge
OnlineBackup_Yes

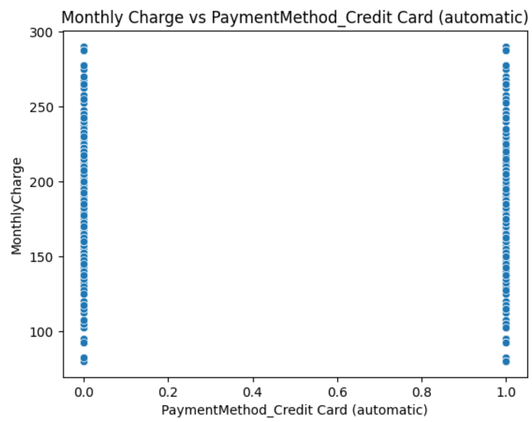

Monthly Charge vs DeviceProtection_Yes
MonthlyCharge
DeviceProtection_Yes

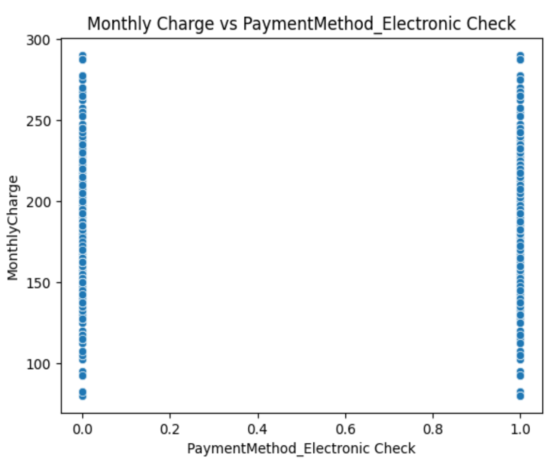

Monthly Charge vs TechSupport_Yes
MonthlyCharge
TechSupport_Yes

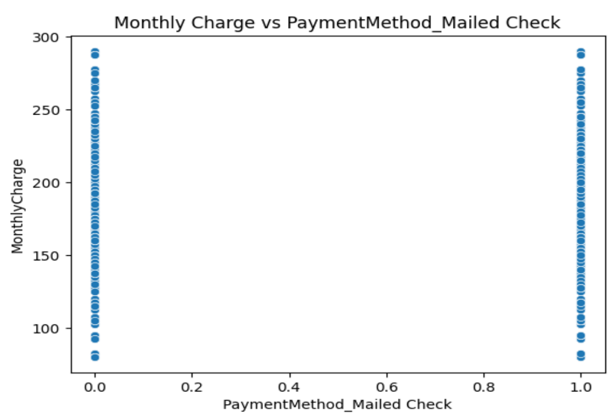

Monthly Charge vs StreamingTV_Yes
MonthlyCharge
StreamingTV_Yes

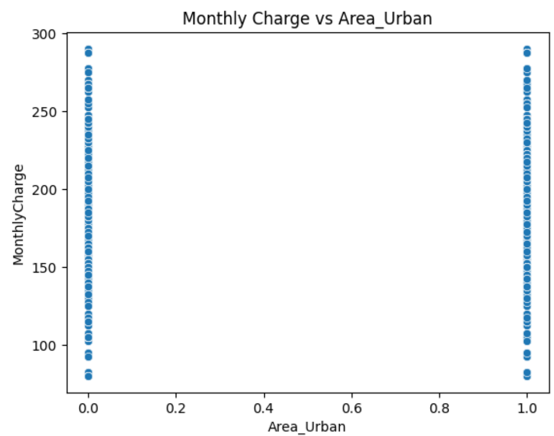

Monthly Charge vs StreamingMovies_Yes
MonthlyCharge
StreamingMovies_Yes

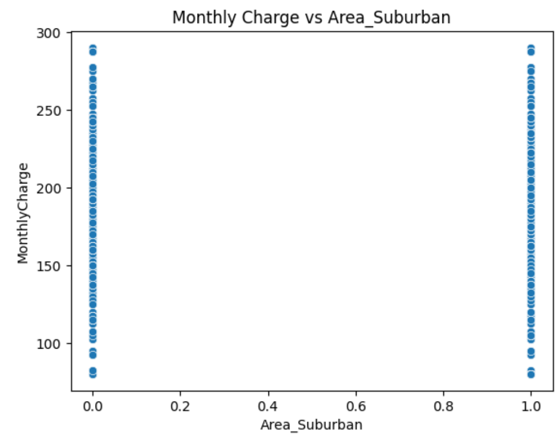

Monthly Charge vs PaperlessBilling_Yes
MonthlyCharge
PaperlessBilling_Yes


Monthly Charge vs PaymentMethod_Credit Card (automatic)
MonthlyCharge
PaymentMethod_Credit Card (automatic)


Monthly Charge vs PaymentMethod_Electronic Check
MonthlyCharge
PaymentMethod_Electronic Check


Monthly Charge vs PaymentMethod_Mailed Check
MonthlyCharge
PaymentMethod_Mailed Check





C4: DATA TRANSFORMATION

The data transformation involves dropping variables with high cardinality since all categorical variables will be covered with numerical variables. This is because many variables can lead to overfitting of the data. Linear regression models also require that the independent variables have numeric data. The Categorical data will be converted to numerical data through dummy encoding (One Hot encoding). Dummy encoding will be used since dummy encoding typically drops one of the binary columns to prevent multicollinearity by setting the option drop_first as true.

C5: PREPARED DATA SET

Churn_df_encoded.csv

D1: INITIAL MODEL

```
INITIAL MULTIPLE LINEAR REGRESSION MODEL
                           Results: Ordinary least squares
=====================================================================================
Model:                     OLS                    Adj. R-squared:          0.946
Dependent Variable:        MonthlyCharge          AIC:                     74402.0277
Date:                      2024-11-17 00:31       BIC:                     74697.6517
No. Observations:          10000                  Log-Likelihood:          -37160.
Df Model:                  40                     F-statistic:             4393.
Df Residuals:              9959                   Prob (F-statistic):      0.00
R-squared:                 0.946                  Scale:                   99.309
-------------------------------------------------------------------------------------
                                       Coef.  Std.Err.    t     P>|t|   [0.025  0.975]
-------------------------------------------------------------------------------------
const                                78.9768   1.1959  66.0392 0.0000 76.6326 81.3210
Population                            0.0000   0.0000   0.4438 0.6572 -0.0000  0.0000
Children                             -0.0017   0.0465  -0.0363 0.9710 -0.0928  0.0895
Age                                   0.0021   0.0048   0.4339 0.6644 -0.0074  0.0116
Income                                0.0000   0.0000   1.4626 0.1436 -0.0000  0.0000
Outage_sec_perweek                   -0.0315   0.0336  -0.9396 0.3474 -0.0973  0.0343
Email                                 0.0005   0.0330   0.0146 0.9883 -0.0642  0.0652
Contacts                              0.0410   0.1010   0.4061 0.6847 -0.1570  0.2390
Yearly_equip_failure                 -0.0842   0.1570  -0.5366 0.5916 -0.3920  0.2235
Replacements                         -0.0514   0.1058  -0.4855 0.6273 -0.2587  0.1560
Reliability                           0.0437   0.1097   0.3987 0.6901 -0.1712  0.2587
Options                               0.1290   0.1131   1.1398 0.2544 -0.0928  0.3508
Respectfulness                        0.0103   0.1136   0.0907 0.9277 -0.2124  0.2330
Courteous                             0.0235   0.1093   0.2148 0.8299 -0.1908  0.2377
Listening                             0.0735   0.1044   0.7040 0.4815 -0.1312  0.2782
Area_Suburban                         0.0969   0.2442   0.3969 0.6914 -0.3817  0.5756
Area_Urban                            0.0095   0.2447   0.0390 0.9689 -0.4701  0.4892
Marital_Married                      -0.0806   0.3159  -0.2550 0.7987 -0.6998  0.5387
Marital_Never Married                -0.2749   0.3140  -0.8755 0.3813 -0.8905  0.3406
Marital_Separated                    -0.1656   0.3115  -0.5318 0.5949 -0.7762  0.4449
Marital_Widowed                      -0.4642   0.3114  -1.4910 0.1360 -1.0745  0.1461
Gender_Male                          -0.1766   0.2023  -0.8733 0.3825 -0.5731  0.2198
Gender_Nonbinary                     -0.8748   0.6716  -1.3026 0.1928 -2.1913  0.4417
Techie_Yes                            0.5128   0.2671   1.9201 0.0549 -0.0107  1.0363
Contract_One year                     0.1546   0.2563   0.6031 0.5465 -0.3478  0.6570
Contract_Two Year                     0.2410   0.2430   0.9915 0.3215 -0.2354  0.7174
Port_modem_Yes                       -0.2339   0.1996  -1.1716 0.2414 -0.6251  0.1574
Tablet_Yes                           -0.2052   0.2182  -0.9403 0.3471 -0.6329  0.2226
InternetService_Fiber Optic          24.7480   0.2010 123.0954 0.0000 24.3539 25.1421
Phone_Yes                            -0.6546   0.3433  -1.9069 0.0566 -1.3276  0.0183
Multiple_Yes                         32.7976   0.2003 163.7787 0.0000 32.4050 33.1901
OnlineSecurity_Yes                    2.8111   0.2084  13.4899 0.0000  2.4026  3.2195
OnlineBackup_Yes                     22.5794   0.2006 112.5725 0.0000 22.1863 22.9726
DeviceProtection_Yes                 12.4275   0.2014  61.7202 0.0000 12.0328 12.8222
TechSupport_Yes                      12.6341   0.2063  61.2437 0.0000 12.2297 13.0385
StreamingTV_Yes                      42.1845   0.1996 211.3041 0.0000 41.7932 42.5758
StreamingMovies_Yes                  52.3478   0.1996 262.2013 0.0000 51.9564 52.7391
PaperlessBilling_Yes                  0.0940   0.2029   0.4634 0.6431 -0.3038  0.4918
PaymentMethod_Credit Card (automatic) -0.3862  0.3041  -1.2699 0.2042 -0.9824  0.2099
PaymentMethod_Electronic Check       -0.2159   0.2721  -0.7936 0.4274 -0.7492  0.3174
PaymentMethod_Mailed Check           -0.0861   0.2970  -0.2900 0.7719 -0.6684  0.4961
-------------------------------------------------------------------------------------
Omnibus:                   27078.757          Durbin-Watson:              2.007
Prob(Omnibus):             0.000              Jarque-Bera (JB):           695.115
Skew:                      -0.051             Prob(JB):                   0.000
Kurtosis:                  1.712              Condition No.:              600547
=====================================================================================
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
[2] The condition number is large, 6.01e+05. This might indicate                that
there are strong multicollinearity or other numerical               problems.
```

D2: JUSTIFICATION OF MODEL REDUCTION

In determining the features to drop, the features that have a p-value >0.05. The p-value represents the probability of observing something. A high p-value indicates that the results are not statistically significant.

According to Beers, Brian (2024, November 13), A p-value, or probability value, is a number describing the likelihood of obtaining the observed data under the null hypothesis of a statistical test. The p-value is an alternative to rejection points to provide the smallest significance level at which the null hypothesis would be rejected. A smaller p-value means stronger evidence in favor of the alternative hypothesis.

The independent features below will be dropped since they have a high p-value, which means they are not statistically significant in determining the monthly charges.

1. Population with a p-value of 0.657.
2. Children with a p-value of 0.971.
3. Age with a p-value of 0.664.
4. Income with a p-value of 0.144.
5. Outage_sec_perweek with a p-value of 0.347.
6. Email with a p-value of 0.988.
7. Contacts with a p-value of 0.685.
8. Yearly_equip_failure with a p-value of 0.592.
9. Replacements with a p-value of 0.627.
10. Reliability with a p-value of 0.690.
11. Options with a p-value of 0.254.
12. Respectfulness with a p-value of 0.928.

13. Courteous with a p-value of 0.830.
14. Listening with a p-value of 0.481.
15. Area_Suburban with a p-value of 0.691.
16. Area_Urban with a p-value of 0.969.
17. Marital_Married with a p-value of 0.799.
18. Marital_Never Married with a p-value of 0.381.
19. Marital_Separated with a p-value of 0.595.
20. Marital_Widowed with a p-value of 0.136.
21. Gender_Male with a p-value of 0.383.
22. Gender_Nonbinary with a p-value of 0.193.
23. Techie_Yes with a p-value of 0.055.
24. Contract_One year with a p-value of 0.546.
25. Contract_Two Year with a p-value of 0.321.
26. Port_modem_Yes with a p-value of 0.241.
27. Tablet_Yes with a p-value of 0.347.
28. Phone_Yes with a p-value of 0.057.
29. PaperlessBilling_Yes with a p-value of 0.643.
30. PaymentMethod_Credit Card (automatic) with a p-value of 0.204.
31. PaymentMethod_Electronic Check with a p-value of 0.427.
32. PaymentMethod_Mailed Check with a p-value of 0.772.

```
print("MODEL REDUCTION")
#Feature reduction using p_value
# drop all columns from model where p-value > 0.05
equation = Churn_df_encoded_model.summary2().tables[1]
temp_drop = []
for i in equation.itertuples():
    if i[4] > 0.05:
        temp_drop.append(i[0])
        print('Drop {} with p-value of {:.3f}.'.format(i[0],i[4]))
X = pd.DataFrame(X) # reset dataframe
X.drop(temp_drop, axis = 1, inplace=True) # drop
```

```
MODEL REDUCTION
Drop Population with p-value of 0.657.
Drop Children with p-value of 0.971.
Drop Age with p-value of 0.664.
Drop Income with p-value of 0.144.
Drop Outage_sec_perweek with p-value of 0.347.
Drop Email with p-value of 0.988.
Drop Contacts with p-value of 0.685.
Drop Yearly_equip_failure with p-value of 0.592.
Drop Replacements with p-value of 0.627.
Drop Reliability with p-value of 0.690.
Drop Options with p-value of 0.254.
Drop Respectfulness with p-value of 0.928.
Drop Courteous with p-value of 0.830.
Drop Listening with p-value of 0.481.
Drop Area_Suburban with p-value of 0.691.
Drop Area_Urban with p-value of 0.969.
Drop Marital_Married with p-value of 0.799.
Drop Marital_Never Married with p-value of 0.381.
Drop Marital_Separated with p-value of 0.595.
Drop Marital_Widowed with p-value of 0.136.
Drop Gender_Male with p-value of 0.383.
Drop Gender_Nonbinary with p-value of 0.193.
Drop Techie_Yes with p-value of 0.055.
Drop Contract_One year with p-value of 0.546.
Drop Contract_Two Year with p-value of 0.321.
Drop Port_modem_Yes with p-value of 0.241.
Drop Tablet_Yes with p-value of 0.347.
Drop Phone_Yes with p-value of 0.057.
Drop PaperlessBilling_Yes with p-value of 0.643.
Drop PaymentMethod_Credit Card (automatic) with p-value of 0.204.
Drop PaymentMethod_Electronic Check with p-value of 0.427.
Drop PaymentMethod_Mailed Check with p-value of 0.772.
```

After performing the model reduction, the Reduced Linear regression model gave the same Adjusted R-squared of 0.946 as the initial model but with fewer variables, indicating that the reduced model is more efficient.

D3: REDUCED LINEAR REGRESSION MODEL

```
REDUCED LINEAR REGRESSION MODEL
FINAL MODEL
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 8 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   InternetService_Fiber Optic   10000 non-null  int64
 1   Multiple_Yes                  10000 non-null  int64
 2   OnlineSecurity_Yes            10000 non-null  int64
 3   OnlineBackup_Yes              10000 non-null  int64
 4   DeviceProtection_Yes          10000 non-null  int64
 5   TechSupport_Yes               10000 non-null  int64
 6   StreamingTV_Yes               10000 non-null  int64
 7   StreamingMovies_Yes           10000 non-null  int64
dtypes: int64(8)
memory usage: 625.1 KB
                      Results: Ordinary least squares
============================================================================
Model:                  OLS                  Adj. R-squared:       0.946
Dependent Variable:     MonthlyCharge        AIC:                  74361.6344
Date:                   2024-11-17 22:22     BIC:                  74426.5275
No. Observations:       10000                Log-Likelihood:       -37172.
Df Model:               8                    F-statistic:          2.198e+04
Df Residuals:           9991                 Prob (F-statistic):   0.00
R-squared:              0.946                Scale:                99.225
----------------------------------------------------------------------------
                             Coef.  Std.Err.    t     P>|t|   [0.025  0.975]
----------------------------------------------------------------------------
const                       78.8429   0.2689 293.2430 0.0000 78.3159 79.3700
InternetService_Fiber Optic 24.7418   0.2007 123.2522 0.0000 24.3483 25.1353
Multiple_Yes                32.7986   0.1999 164.1060 0.0000 32.4068 33.1904
OnlineSecurity_Yes           2.8009   0.2079  13.4720 0.0000  2.3934  3.2084
OnlineBackup_Yes            22.5755   0.2002 112.7469 0.0000 22.1830 22.9680
DeviceProtection_Yes        12.4560   0.2008  62.0275 0.0000 12.0624 12.8496
TechSupport_Yes             12.6418   0.2059  61.4057 0.0000 12.2382 13.0454
StreamingTV_Yes             42.1795   0.1993 211.6736 0.0000 41.7889 42.5701
StreamingMovies_Yes         52.3391   0.1994 262.5468 0.0000 51.9484 52.7299
----------------------------------------------------------------------------
Omnibus:                 30980.596          Durbin-Watson:             2.008
Prob(Omnibus):           0.000              Jarque-Bera (JB):          702.004
Skew:                    -0.052             Prob(JB):                  0.000
Kurtosis:                1.706              Condition No.:             5
============================================================================
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is
correctly specified.
```

E1: MODEL COMPARISON

- a model evaluation metric

1. Durbin-Watson Statistic:

Initial model: 2.007

Reduced linear regression model: 2.008

The Durbin-Watson statistic tests for autocorrelation in the residuals. Values closer to 2 indicate no autocorrelation. Both models are very close to 2, showing no significant autocorrelation.

2. Jarque-Bera Test (JB):

   Initial Model: 695.115

   Reduced linear regression model: 702.004

   The JB test examines normality based on skewness and kurtosis. The higher the JB value, the more likely the residuals deviate from normality. The reduced linear regression model has a slightly higher JB value, indicating it may have slightly more non-normality than the initial model.

3. Skew:

Initial model: -0.051

Reduced linear regression model: -0.052

   Both models show a similar skewness close to 0, meaning the residuals are nearly symmetric. The values are virtually identical, with the Reduced linear regression model having an almost identical skew.

4. Kurtosis:

Initial model: 1.712

Reduced linear regression model: 1.706

The Kurtosis measures the "tailed Ness" of the distribution. A value of 3 indicates normality, while values less than 3 suggest light tails. Both models have low kurtosis, implying light tails, but the difference is minimal, with the initial model showing a slightly higher kurtosis Reduced linear regression model.

5. Condition Number:

Initial model: 600547

Reduced linear regression model: 5

The condition number indicates potential multicollinearity or instability in the initial model. A higher condition number in the initial model signals potential problems with multicollinearity, which is not present in the reduced linear regression model with a very low condition number.

6. Omnibus Test:

Initial model 27078.757

Reduced linear regression model:30980.596

The Omnibus test checks for deviations from normality in the residuals (skewness and kurtosis). A higher value indicates a greater deviation from normality. The reduced linear regression model has a higher Omnibus value, suggesting slightly more deviation from normality than the initial model.

7. R squared and Adjusted R squared

Initial model: R squared- 0.946, Adjusted R squared – 0.946

Reduced linear regression model: R squared- 0.946, Adjusted R squared – 0.946

The R squared value for both the initial multiple linear regression model and reduced linear regression model is 0.946, and the Adjusted R squared is also 0.946 however, the reduced linear regression model has achieved the same adjusted R squared rate using 8 predictor variables with a p-value of less than 0.05 hence indicating that there are more significant, whereas the initial multiple linear regression model has 40 predictor variables.

8. MSE and RMSE

   Initial model: MSE- 98.901 ,RMSE-9.944

   Reduced linear regression model: MSE- 99.135 , RMSE-9.956

   The MSE (Mean Squared Error) equals 99.135 and the RMSE equals 9.956 (Root Mean Squared Error). The RMSE is relatively low for the reduced linear regression, which performs well. On the other hand, the MSE for the initial model is 98.901, and the RMSE is 9.944. The difference in RMSE is very small, suggesting that both models perform similarly in terms of the magnitude of the error in the prediction

9. Akaike Information Criterion (AIC)

   Reduced Model: 74361.6344

   Initial Model: 74402.0277

The reduced model has a slightly lower AIC, which suggests it fits better than the initial model. A lower AIC indicates a more optimal model balancing goodness of fit and model complexity.

10. Bayesian Information Criterion (BIC)

Reduced Model: 74426.5275

Initial Model: 74697.6517

The reduced model also has a lower BIC. Like AIC, BIC penalizes the number of predictors, and a lower value indicates a more parsimonious model with a better fit. The reduced model performs better in this regard as well.

11. F-statistic

Reduced Model: 21980

Initial Model: 4393

The F-statistic measures the overall significance of the model. A higher F-statistic suggests a better fit. The reduced model has a much higher F-statistic, indicating that it explains the variance in the dependent variable more efficiently despite having fewer predictors.

12. Degrees of Freedom (Df Model)

Reduced Model: 8

Initial Model: 40

The reduced model has only 8 predictors, while the initial model has 40. This indicates that the reduced model is much simpler.

13. Log-Likelihood:

    Reduced Model: -37172

    Initial Model: -37160

    The log-likelihood values are very similar, but the reduced model has a slightly worse log-likelihood, although it still fits the data well regarding predictive power.

14. Residuals Scale:

    Reduced Model: 99.225

    Initial Model: 99.309

    Both models have very similar residual scales, indicating that their predictions are nearly identical regarding variance.

15. Prob (F-statistic):

    Reduced Model: 0.00

    Initial Model: 0.00

    The very low p-value for both models suggests that both models are highly statistically significant, with a negligible chance of the results occurring randomly.

```
    print('   {:+.3f} x ( {} ) '.format(i[1],i[0]))
```

```
Adj. R-squared: 0.946
Estimate [MonthlyCharge] as y =
   +78.843 x ( const )
   +24.742 x ( InternetService_Fiber Optic )
   +32.799 x ( Multiple_Yes )
   +2.801 x ( OnlineSecurity_Yes )
   +22.575 x ( OnlineBackup_Yes )
   +12.456 x ( DeviceProtection_Yes )
   +12.642 x ( TechSupport_Yes )
   +42.179 x ( StreamingTV_Yes )
   +52.339 x ( StreamingMovies_Yes )
```

In [84]:
```
#Comments : The Adjusted R squared is 94.6 meaning the model fits the data by 94.6%
```

In [100]:
```
# Calculating MSE and RMSE
def mse(actual, pred):
    actual, pred = np.array(actual), np.array(pred)
    return np.square(np.subtract(actual, pred)).mean()

def rmse(actual, pred):
    return np.sqrt(mse(actual, pred))

# Assuming `y` is the actual target variable (e.g., MonthlyCharge)
# and `Churn_df_encoded_model` is the fitted model from statsmodels

pred = Churn_df_encoded_model.predict(Xc)

# Calculate MSE and RMSE
mse_value = mse(y, pred)
rmse_value = rmse(y, pred)

# Print the results
print(f"MSE: {mse_value}")
print(f"RMSE: {rmse_value}")
```

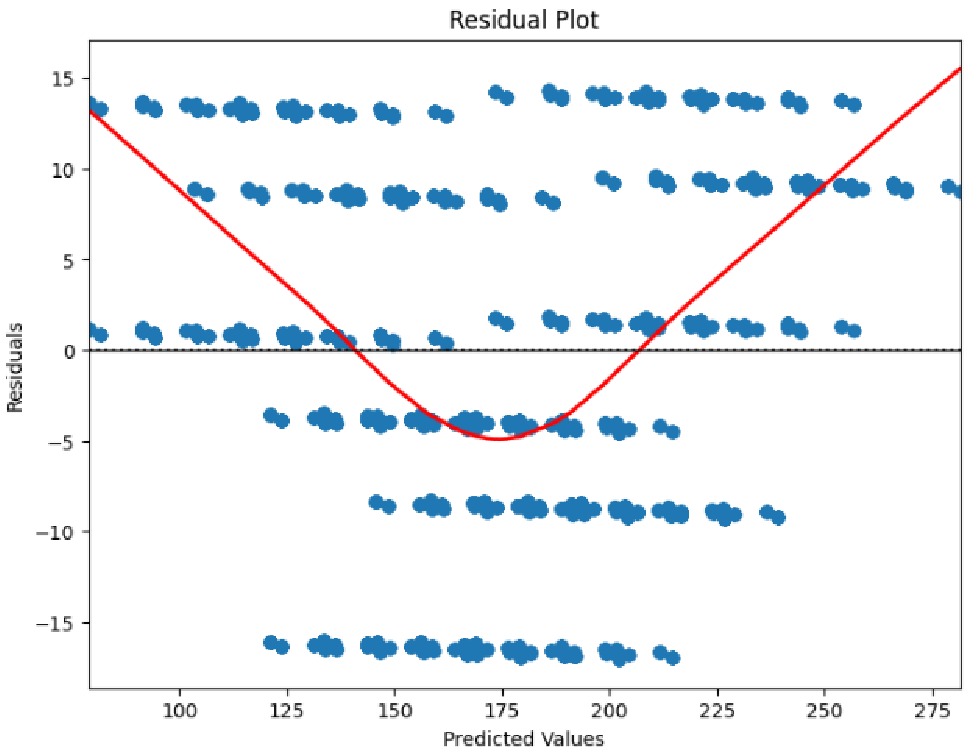
```
MSE: 99.13537867371001
RMSE: 9.956675081256293
```

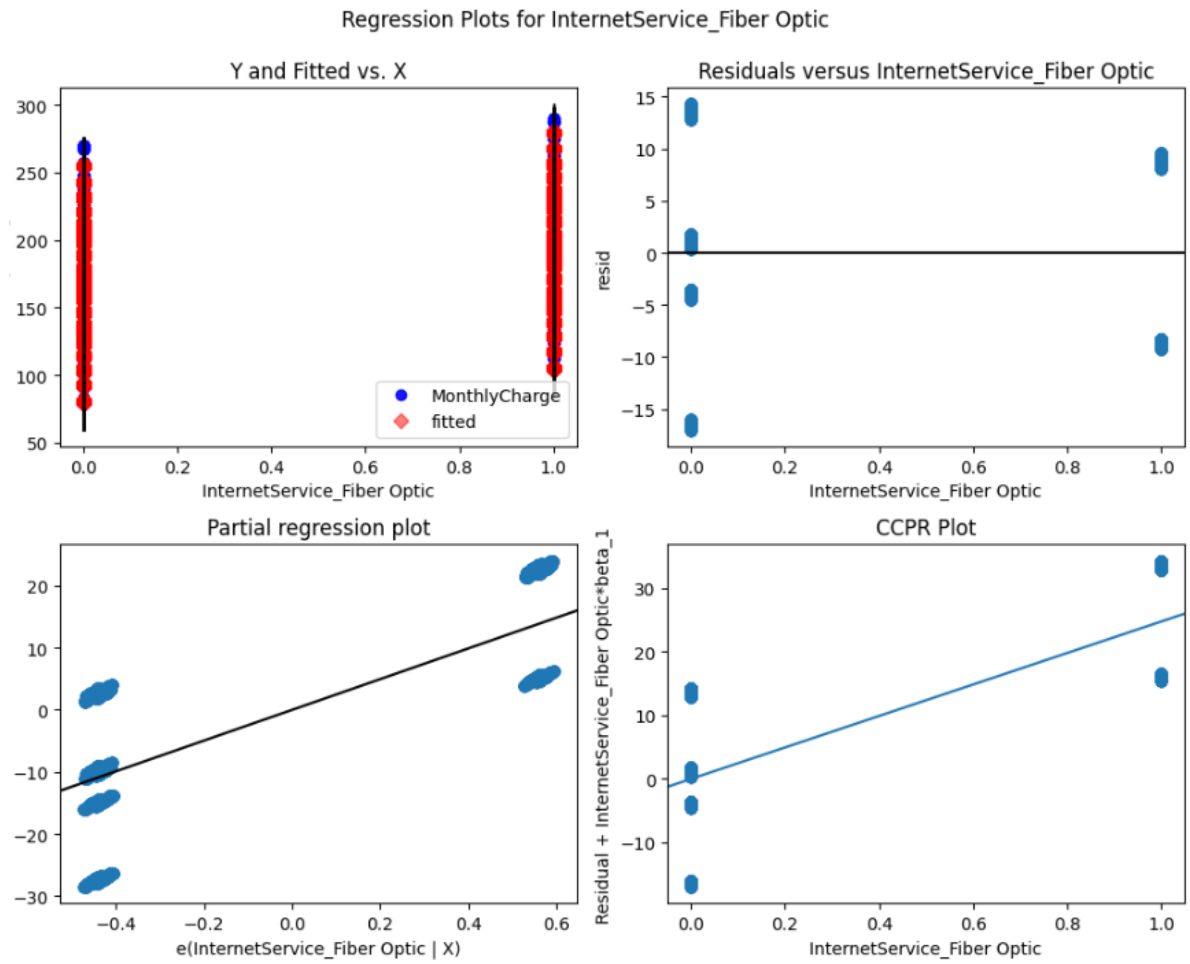
E2: OUTPUT AND CALCULATIONS

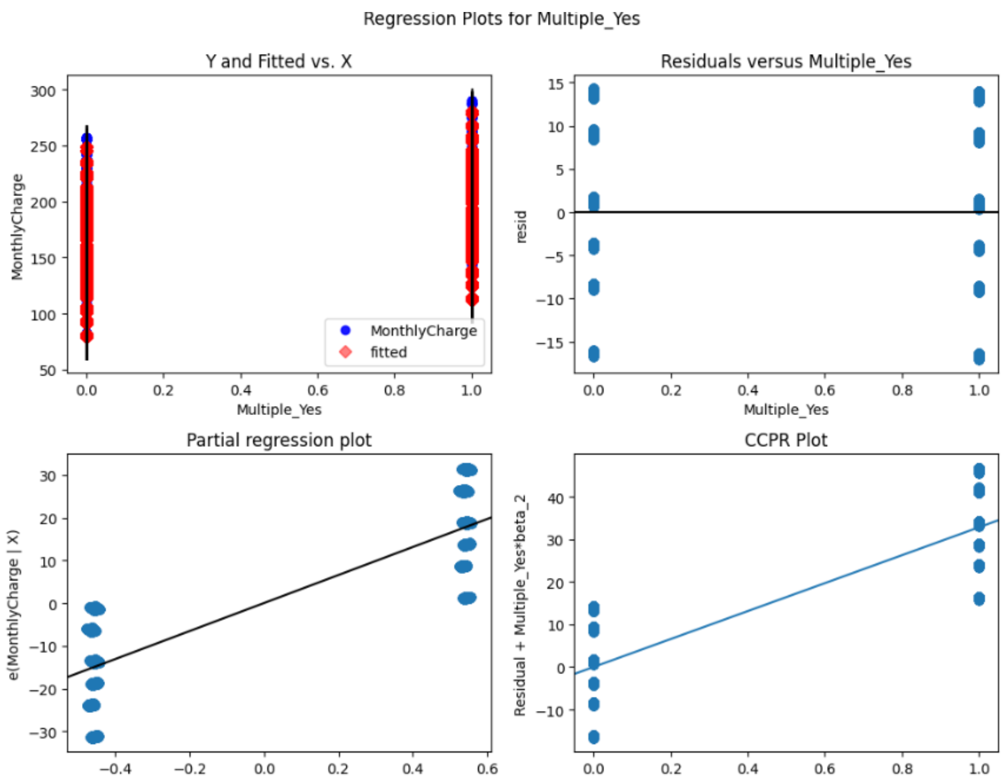
Provide the output and *all* calculations of the analysis you performed, including the following elements for your reduced linear regression model:

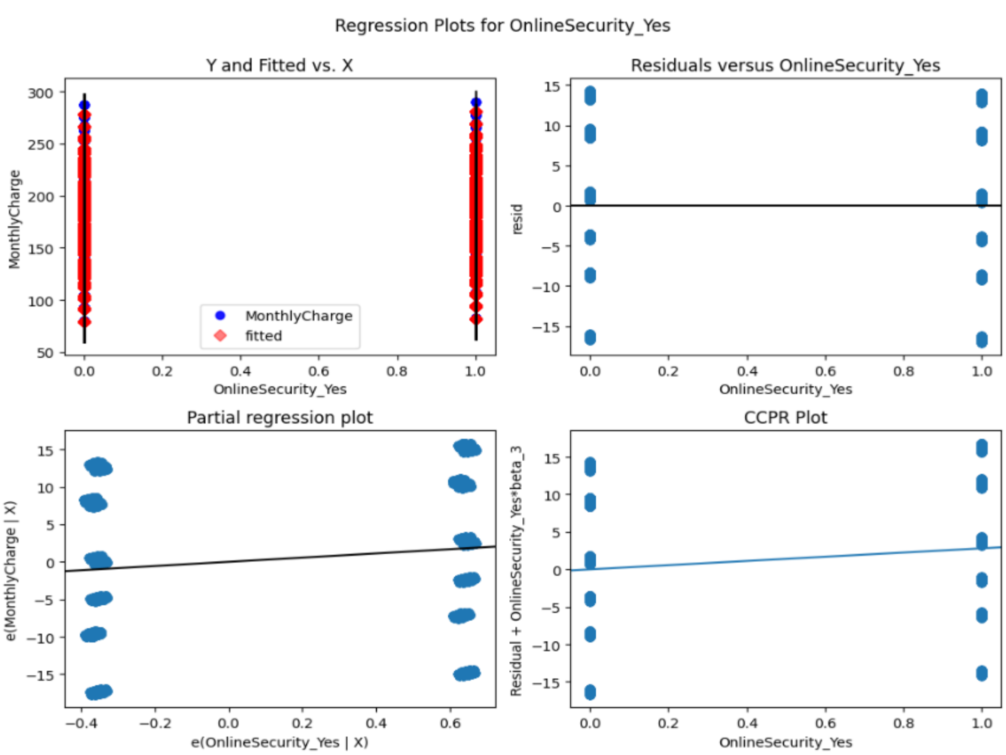
- a residual plot

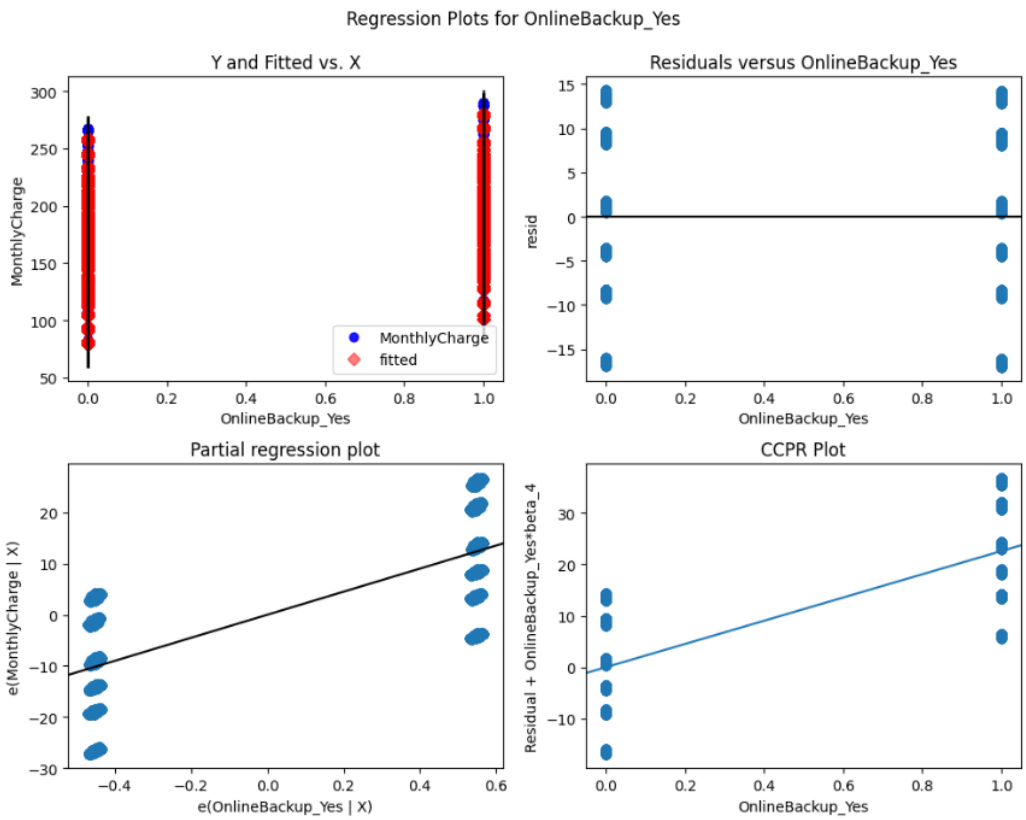
```python
# Create the residual plot
predicted = Churn_df_encoded_model.predict(Xc)
residuals = Churn_df_encoded_model.resid
plt.figure(figsize=(8, 6))
sns.residplot(x=predicted, y=residuals, lowess=True, line_kws={'color': 'red', 'lw': 2})
plt.axhline(0, color='black', linestyle='-', lw=1)
plt.title("Residual Plot")
plt.xlabel("Predicted Values")
plt.ylabel("Residuals")
plt.show()
```

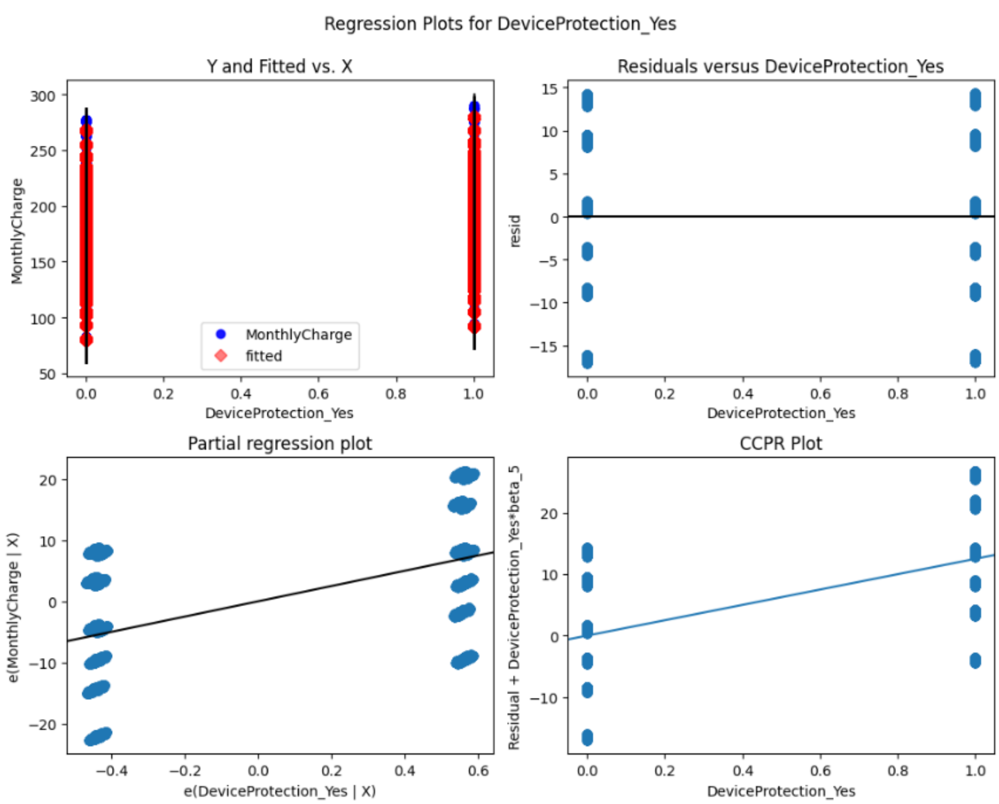


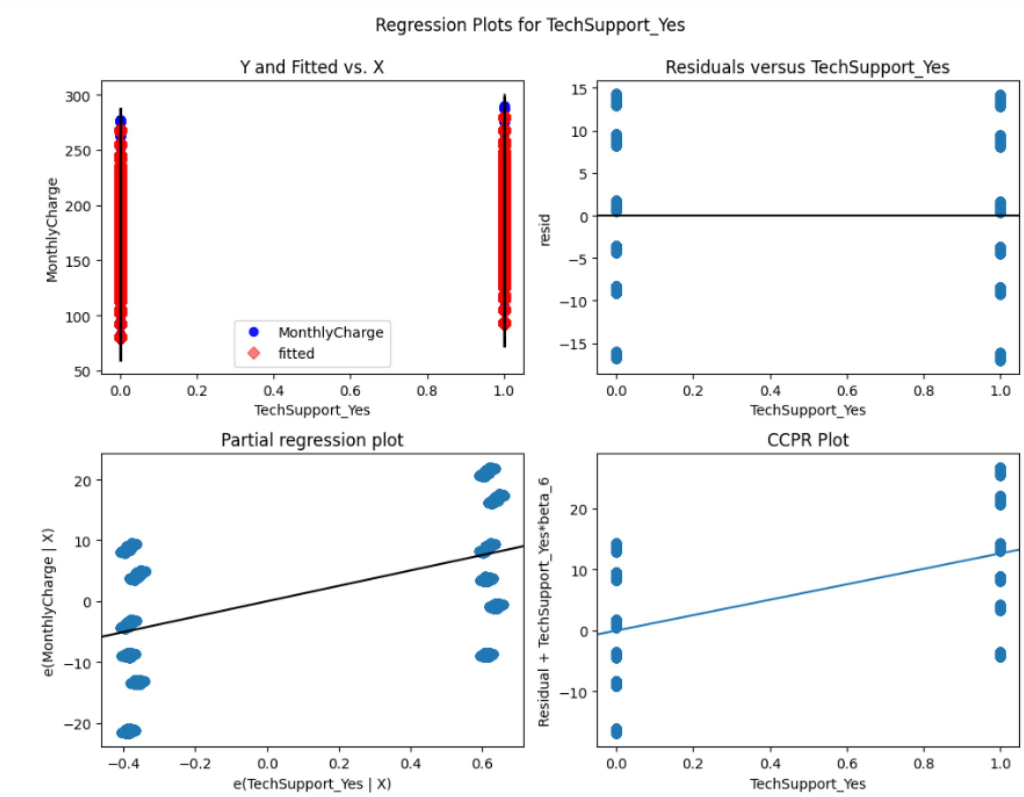

Regression Plots for InternetService_Fiber Optic
Y and Fitted vs. X
MonthlyCharge
fitted
InternetService_Fiber Optic
Residuals versus InternetService_Fiber Optic
resid
InternetService_Fiber Optic
Partial regression plot
e(InternetService_Fiber Optic | X)
CCPR Plot
Residual + InternetService_Fiber Optic*beta_1
InternetService_Fiber Optic

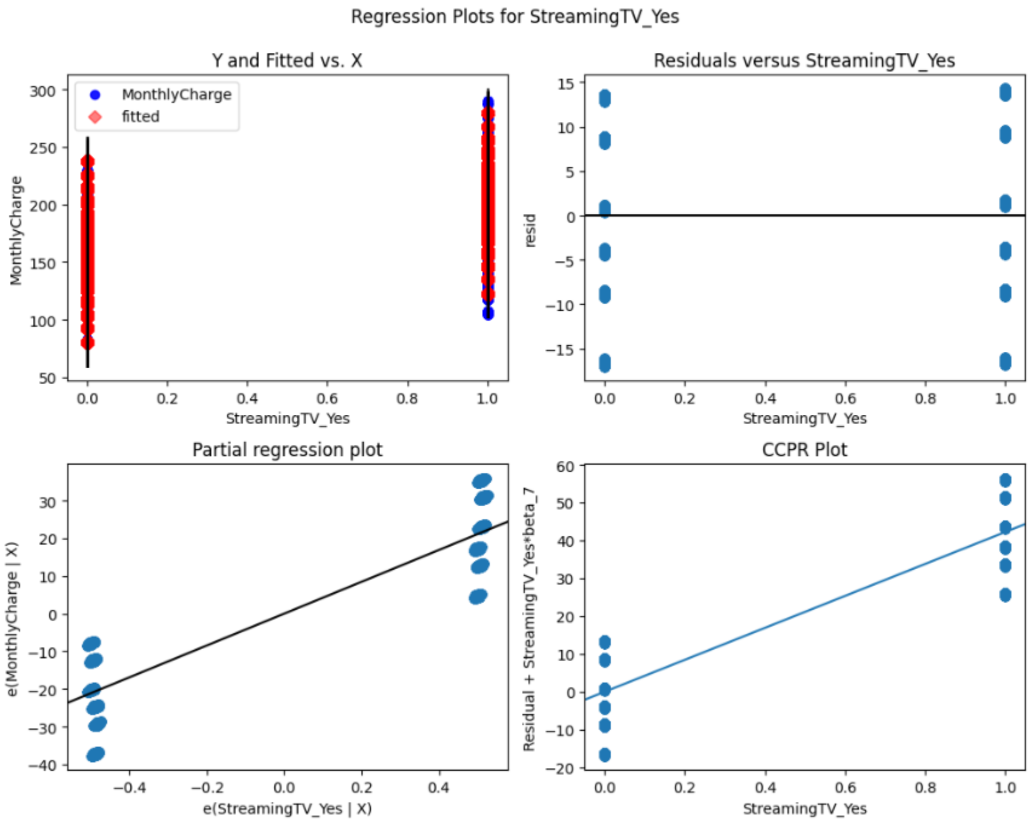

Regression Plots for Multiple_Yes
Y and Fitted vs. X
MonthlyCharge
Multiple_Yes
MonthlyCharge
fitted
Residuals versus Multiple_Yes
resid
Multiple_Yes
Partial regression plot
e(MonthlyCharge | X)
CCPR Plot
Residual + Multiple_Yes*beta_2

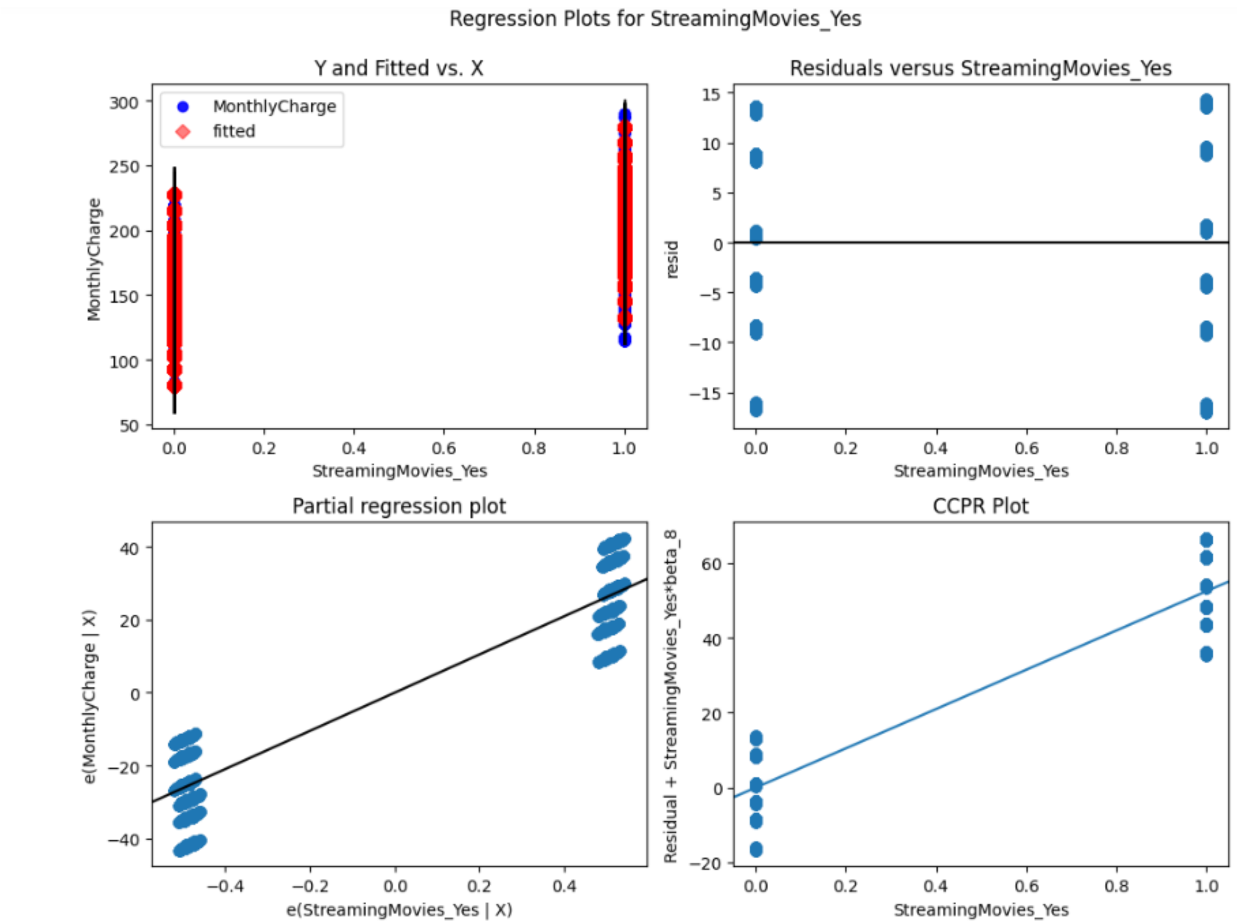

Regression Plots for OnlineSecurity_Yes
Y and Fitted vs. X
MonthlyCharge
OnlineSecurity_Yes
MonthlyCharge
fitted
Residuals versus OnlineSecurity_Yes
resid
OnlineSecurity_Yes
Partial regression plot
e(MonthlyCharge | X)
e(OnlineSecurity_Yes | X)
CCPR Plot
Residual + OnlineSecurity_Yes*beta_3
OnlineSecurity_Yes


Regression Plots for OnlineBackup_Yes
Y and Fitted vs. X
MonthlyCharge
fitted
OnlineBackup_Yes
Residuals versus OnlineBackup_Yes
resid
Partial regression plot
e(MonthlyCharge | X)
e(OnlineBackup_Yes | X)
CCPR Plot
Residual + OnlineBackup_Yes*beta_4


Regression Plots for DeviceProtection_Yes
Y and Fitted vs. X
MonthlyCharge
fitted
DeviceProtection_Yes
Residuals versus DeviceProtection_Yes
resid
Partial regression plot
e(MonthlyCharge | X)
e(DeviceProtection_Yes | X)
CCPR Plot
Residual + DeviceProtection_Yes*beta_5


Regression Plots for TechSupport_Yes
Y and Fitted vs. X
MonthlyCharge
fitted
TechSupport_Yes
Residuals versus TechSupport_Yes
resid
Partial regression plot
e(MonthlyCharge | X)
e(TechSupport_Yes | X)
CCPR Plot
Residual + TechSupport_Yes*beta_6


Regression Plots for StreamingTV_Yes
Y and Fitted vs. X
MonthlyCharge
fitted
MonthlyCharge
StreamingTV_Yes
Residuals versus StreamingTV_Yes
resid
StreamingTV_Yes
Partial regression plot
e(MonthlyCharge | X)
e(StreamingTV_Yes | X)
CCPR Plot
Residual + StreamingTV_Yes*beta_7
StreamingTV_Yes



- **the model's residual standard error**

```
#RESIDUAL STANDARD ERROR
residuals=Churn_df_encoded_model.resid
RSS = sum(residuals**2)
n = len(y)
p = len(Churn_df_encoded_model.params)
RSE = np.sqrt(RSS / (n - p))
print(f"Residual Standard Error: {RSE:.2f}")
```

```
Residual Standard Error: 9.96
```

```
                       Results: Ordinary least squares
============================================================================
Model:                   OLS                 Adj. R-squared:        0.946
Dependent Variable:      MonthlyCharge       AIC:                   74361.6344
Date:                    2024-11-17 22:22    BIC:                   74426.5275
No. Observations:        10000               Log-Likelihood:        -37172.
Df Model:                8                   F-statistic:           2.198e+04
Df Residuals:            9991                Prob (F-statistic):    0.00
R-squared:               0.946               Scale:                 99.225
----------------------------------------------------------------------------
                             Coef.  Std.Err.    t     P>|t|   [0.025  0.975]
----------------------------------------------------------------------------
const                       78.8429   0.2689 293.2430 0.0000 78.3159 79.3700
InternetService_Fiber Optic 24.7418   0.2007 123.2522 0.0000 24.3483 25.1353
Multiple_Yes                32.7986   0.1999 164.1060 0.0000 32.4068 33.1904
OnlineSecurity_Yes           2.8009   0.2079  13.4720 0.0000  2.3934  3.2084
OnlineBackup_Yes            22.5755   0.2002 112.7469 0.0000 22.1830 22.9680
DeviceProtection_Yes        12.4560   0.2008  62.0275 0.0000 12.0624 12.8496
TechSupport_Yes             12.6418   0.2059  61.4057 0.0000 12.2382 13.0454
StreamingTV_Yes             42.1795   0.1993 211.6736 0.0000 41.7889 42.5701
StreamingMovies_Yes         52.3391   0.1994 262.5468 0.0000 51.9484 52.7299
----------------------------------------------------------------------------
Omnibus:                   30980.596         Durbin-Watson:            2.008
Prob(Omnibus):             0.000             Jarque-Bera (JB):         702.004
Skew:                      -0.052            Prob(JB):                 0.000
Kurtosis:                  1.706             Condition No.:            5
============================================================================
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is
correctly specified.
```

E3: CODE

Please modify the input and output file path

InputFilePath = 'C:/Users/beatr/Downloads/churn_clean.csv'

Outputfile='C:/Users/beatr/Downloads/Churn_df_encoded.csv'

BeatriceMwangi_D2
08_DataModellinf_T

F1: RESULTS

1. **Discuss the results of your data analysis, including the following elements:**

- **a regression equation for the reduced model**

```
# equation of the regression line/plane
print('Adj. R-squared: {}'.format(Churn_df_encoded_model.summary2().tables[0][3][0]))
equation = Churn_df_encoded_model.summary2().tables[1]
print('Estimate [{}] as y = '.format(Churn_df_encoded_model.summary2().tables[0][1][1]))
for i in equation.itertuples():
    print('    {:+.3f} x ( {} ) '.format(i[1],i[0]))
```

```
Adj. R-squared: 0.946
Estimate [MonthlyCharge] as y =
   +78.843 x ( const )
   +24.742 x ( InternetService_Fiber Optic )
   +32.799 x ( Multiple_Yes )
   +2.801 x ( OnlineSecurity_Yes )
   +22.575 x ( OnlineBackup_Yes )
   +12.456 x ( DeviceProtection_Yes )
   +12.642 x ( TechSupport_Yes )
   +42.179 x ( StreamingTV_Yes )
   +52.339 x ( StreamingMovies_Yes )
```

The regression equation for the reduced model is

[MonthlyCharge] as y = +78.843 +24.742 (InternetService_Fiber Optic) +32.799 (Multiple_Yes) +2.801 (OnlineSecurity_Yes) +22.575 (OnlineBackup_Yes) +12.456 (DeviceProtection_Yes) +12.642 (TechSupport_Yes) +42.179 (StreamingTV_Yes) +52.339 (StreamingMovies_Yes)

- **an interpretation of the coefficients of the reduced model**

The Adjusted R squared is 94.6, meaning the model fits the data by 94.6%. In determining the monthly charge, keeping all things constant, a customer enrolled for basic service in the telecommunication will pay 78.843 if the customer has internet service -fiber optics, the monthly charge will go up by 24.742 if they take up multiple services, the monthly charge will go up 32.799 if they add the online security, the monthly charge will increase by 2.801, if they add the OnlineBackup, the monthly charge will increase by 22.575, if they add the device protection, the monthly charge will increase by 12.546, if they add the Tech Support the monthly charge will increase by 12.642, if they add the Streaming TV, the monthly charge will increase by 42.179 if they add the Streaming Movies, the monthly charge will increase by 52.339. With this equation, the monthly charge can be accurately predicted by 94. 6 percent based on the adjusted R squared.

- **the statistical and practical significance of the reduced model**

```
                      Results: Ordinary least squares
===========================================================================
Model:                 OLS                Adj. R-squared:       0.946
Dependent Variable:    MonthlyCharge      AIC:                  74361.6344
Date:                  2024-11-16 16:53   BIC:                  74426.5275
No. Observations:      10000              Log-Likelihood:       -37172.
Df Model:              8                  F-statistic:          2.198e+04
Df Residuals:          9991               Prob (F-statistic):   0.00
R-squared:             0.946              Scale:                99.225
---------------------------------------------------------------------------
                            Coef.  Std.Err.    t     P>|t|   [0.025  0.975]
---------------------------------------------------------------------------
const                      78.8429  0.2689 293.2430 0.0000 78.3159 79.3700
InternetService_Fiber Optic 24.7418  0.2007 123.2522 0.0000 24.3483 25.1353
Multiple_Yes               32.7986  0.1999 164.1060 0.0000 32.4068 33.1904
OnlineSecurity_Yes          2.8009  0.2079  13.4720 0.0000  2.3934  3.2084
OnlineBackup_Yes           22.5755  0.2002 112.7469 0.0000 22.1830 22.9680
DeviceProtection_Yes       12.4560  0.2008  62.0275 0.0000 12.0624 12.8496
TechSupport_Yes            12.6418  0.2059  61.4057 0.0000 12.2382 13.0454
StreamingTV_Yes            42.1795  0.1993 211.6736 0.0000 41.7889 42.5701
StreamingMovies_Yes        52.3391  0.1994 262.5468 0.0000 51.9484 52.7299
---------------------------------------------------------------------------
Omnibus:                30980.596         Durbin-Watson:           2.008
Prob(Omnibus):          0.000             Jarque-Bera (JB):        702.004
Skew:                   -0.052            Prob(JB):                0.000
Kurtosis:               1.706             Condition No.:           5
===========================================================================
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is
correctly specified.
```

The output of the ordinary Least Squares (OLS) regression model is indicated below.

R-squared: 0.946: This means that 94.6% of the variation in the dependent variable (MonthlyCharge) is explained by the independent variables(InternetService_Fiber Optic, Multiple_Yes, OnlineSecurity_Yes, OnlineBackup_Yes, DeviceProtection_Yes, TechSupport_Yes, StreamingTV_Yes, StreamingMovies_Yes) in the model. This is a high R-squared value, indicating that the model explains a large portion of the variability in the target variable.

Adjusted R-squared: 0.946: Adjusted R-squared is similar to R-squared but adjusts for the number of predictors in the model. Since the model has multiple predictors, the high Adjusted R-squared value suggests that the model is well-fitting even after accounting for the number of predictors. As such, there is no overfitting.

Coef. (Coefficient): These values represent the estimated effect of each predictor on the dependent variable (MonthlyCharge). For example, for InternetService_Fiber Optic, the coefficient is 24.7418, meaning that, holding all other variables constant, switching from a different Internet service type to "Fiber Optic" would increase the MonthlyCharge by approximately 24.74 units.

The Standard Error(Std.Err.) This measures the accuracy of the coefficient estimate. A smaller standard error indicates a more precise estimate.

t: The t-value tests whether the coefficient is significantly different from zero. Larger t-values indicate that the variable has a stronger effect on the dependent variable.

The p-value(P>|t|) for each coefficient. A p-value less than 0.05 (usually) indicates that the predictor is statistically significant in the model. All the predictor features have p-values of 0.0000, significantly contributing to explaining the Monthly Charge.

Each coefficient has 95% confidence intervals([0.025, 0.975]). For instance, the confidence interval for InternetService_Fiber Optic is [24.3483, 25.1353], which means that, with 95% confidence, the actual coefficient for this predictor lies within this range.

- **the limitations of the data analysis**

In analyzing this data, several limitations were encountered. The analysis was limited due to a lack of understanding of the business data, how the data was collected, and the domain knowledge. More so, the data lacked appropriate dates to explore the effect of various features over a time period.

F2: RECOMMENDATIONS

Based on the results obtained from the linear regression model. The telecommunication company can predict the monthly charge they will get from their customer 94.6 percent of the time. The linear regression model output provides strong evidence that the model is a good fit for predicting the monthly charge; however, more analysis is needed to improve model assumptions and residual behavior.

G: VIDEO

https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=16607f7c-8389-4cbc-945d-b22c006e58eb

H: SOURCES FOR THIRD-PARTY CODE

I: SOURCES

Bobbitt, Zach (2021, November 16) The Five Assumptions of Multiple Linear Regression

https://www.statology.org/multiple-linear-regression-assumptions/

Bhandari, Pritha (2023, June 22) Independent vs. Dependent Variables | Definition & Examples

https://www.scribbr.com/methodology/independent-and-dependent-variables/

Coursera staff (2024, June 27) What Is Linear Regression? (Types, Examples, Careers)

https://www.coursera.org/articles/linear-regression?msockid=27f03d3f26886f0110ac2eff27066ea6

Saxena, Shipra (2024, September 18) What are Categorical Data Encoding Methods | Binary Encoding

https://www.analyticsvidhya.com/blog/2020/08/types-of-categorical-data-encoding/

Beers, Brian (2024, November 13), P-Value: What It Is, How to Calculate It, and Examples

https://www.investopedia.com/terms/p/p-value.asp

J: PROFESSIONAL COMMUNICATION